

claude-windows / e1710074-0d77-443d-9241-d89dcd068626

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\e1710074-0d77-443d-9241-d89dcd068626.jsonl`  
- SHA-256: `0d0e0498c95334c70d3b79b65387707a0d39edb7741eabc4524f09964bc98af4`  
- Source modified: `2026-06-21T00:58:47+00:00`  
- Imported at: `2026-07-05T16:48:19+00:00`  
- project: `C--Users-avidu-Projects-badminton-highlight-indexer`  
- session_id: `e1710074-0d77-443d-9241-d89dcd068626`
```

Transcript

1. user (2026-06-21T00:40:13.244Z)

The Gen-0 weights bundle written by `worker train` records `git\_sha: "unknown"` in `weights/<version>/manifest.json` when the run happens on a remote box ported via `git archive` (the tarball has no `.git`, so `training/gen0/harness.py::\_git\_sha()` falls back to "unknown"). This was observed on the 2026-06-20 GCP L4 M2 run ? `gs://khelsutra-rally-corpus/weights/0.1.0-l4/manifest.json` has no real provenance.

Fix: let the git SHA be supplied out-of-band so the bundle is provenance-stamped even off a tarball. Options: (1) have `\_git\_sha()` in `training/gen0/harness.py` fall back to an env var (e.g. `RALLY\_GIT\_SHA`) when `git rev-parse` fails; (2) embed a `VERSION`/`GIT\_SHA` file when building the deploy tarball (`git archive` + `git rev-parse HEAD` written into the archive) and read it; and/or (3) have `backend/worker` pass the SHA through to the harness `--git-sha`. Add a unit test asserting the manifest carries a real SHA when the env/file is present. Keep it small + default-OFF-safe (don't break the local `git`-present path). Branch off origin/master; one PR.

2. assistant (2026-06-21T00:40:19.411Z)

I'll start by following the repo-freshness convention (pull first) and exploring the relevant code.

3. user (2026-06-21T00:40:24.219Z)

```
From https://github.com/Khelsutra/badminton-highlight-indexer  
a555887..e6110ce master -> origin/master  
Your configuration specifies to merge with the ref 'refs/heads/docs/m2-finalized-report'  
from the remote, but no such ref was fetched.
```

4. user (2026-06-21T00:40:24.263Z)

```
1      """Gen-0 TAL harness (M0.4): ONE command -> train -> honest LOVO eval -> ship-gate ->  
artifact.  
2  
3      python -m training.gen0.harness --manifest output/human_lovo_manifest.json \  
4      --floor eval_baselines/heuristic_lovo_n6.json --out artifacts/rally_tal_gen0  
--version 0.1.0  
5  
6      Productionizes the rally_seq prototype per ADR-008:  
7      - TRAINS on the corpus manifest (per-video trajectory + golden labels), evaluates with  
8      honest leave-one-video-out (split by match, threshold tuned on train only), then fits  
9      the FINAL model on all videos.  
10     - Emits a versioned ARTIFACT BUNDLE ? the only thing serving may ever consume:  
11         <out>/<version>/weights.json      (gitignored; classifier + frozen normalization)  
12         <out>/<version>/manifest.json     (the reviewable contract: feature-schema version,  
13                                         calibration, training lineage + GT hashes,  
14                                         eval results, attestations, gate verdict)  
15     - Runs the SHIP-GATE as pre-flight: floor comparison + a PAIRED per-video sign test
```

```

16         (at n=6 a mean gap alone is noise-ambiguous), plus the hard attestation blocks
17         (WASB C3 unresolved => commercial ship refused; explicit F1@tIoU target still
18         undefined => no self-declared victory). The product side owns final admission
19         (backend/eval/regression.py); this harness never grades itself into serving.
20
21     Training may import backend.* (one-way wall); backend must never import training.*.
22     """
23     from __future__ import annotations
24
25     import argparse
26     import hashlib
27     import json
28     import os
29     import subprocess
30     import time
31     from math import comb
32     from typing import Dict, List, Optional
33
34     import numpy as np
35
36     from backend.eval.classifier import LogisticRegression
37     from backend.eval.gt_loader import load_gt_intervals
38     from backend.eval.regression import gt_hash
39     from backend.eval.rally_seq_proto import (
40         labels_for, run as lovo_run, _seg_f1,
41     )
42     from backend.features.rally_seq import FEAT_NAMES, FEATURE_SCHEMA, build_frame_arrays,
features
43     from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
44
45     # Post-processing constants ? part of the model's identity, recorded in the manifest.
46     CALIBRATION_DEFAULTS = {"smooth": 7, "merge_gap": 1.0, "min_dur": 1.0}
47     THRESHOLD_GRID = [round(t, 2) for t in np.arange(0.2, 0.81, 0.05)]
48
49
50     def _git_sha() -> str:
51         try:
52             return subprocess.run(["git", "rev-parse", "HEAD"], capture_output=True,
53                                   text=True, timeout=10).stdout.strip() or "unknown"
54         except Exception:
55             return "unknown"
56
57
58     def _sha256(path: str) -> str:
59         h = hashlib.sha256()
60         with open(path, "rb") as f:
61             for chunk in iter(lambda: f.read(65536), b''):
62                 h.update(chunk)
63         return h.hexdigest()
64
65
66     def sign_test(learned: List[float], floor: List[float]):
67         """Paired one-sided sign test: is learned > floor more often than chance?
68
69         At n=6 a mean comparison alone is inside noise (SE ~0.11); the paired per-video
70         wins are the defensible statistic. Returns (wins, n, p_one_sided)."""
71         pairs = [(lv, fv) for lv, fv in zip(learned, floor) if lv != fv] # ties drop,
standard
72         wins = sum(1 for lv, fv in pairs if lv > fv)
73         n = len(pairs)
74         p = sum(comb(n, k) for k in range(wins, n + 1)) / (2 ** n) if n else 1.0
75         return wins, n, p
76
77
78     def train_final(specs: List[Dict]):

```

```

79         """Fit the FINAL model on ALL corpus videos; threshold tuned on the same (train-
side).
80
81     The honest generalization estimate is the LOVO result, not this fit ? recorded as
such."""
82     vids = []
83     for s in specs:
84         pts = TrackNetRunner.parse_trajectory_csv(s["trajectory"])
85         arr = build_frame_arrays(pts, float(s["frame_width"]), float(s["fps"]))
86         X, times = features(arr, float(s["fps"]))
87         gts = load_gt_intervals(s["labels"], strict=False)
88         vids.append({"name": s["name"], "X": X, "times": times,
89                     "y": labels_for(times, gts), "gts": gts})
90     Xall = np.vstack([v["X"] for v in vids])
91     yall = np.concatenate([v["y"] for v in vids])
92     clf = LogisticRegression(lr=0.2, epochs=2000, l2=1e-2)
93     clf.fit(Xall, yall, feature_names=FEAT_NAMES)
94     thr = max(THRESHOLD_GRID, key=lambda t: _seg_f1(clf, vids, float(t)))
95     return clf, float(thr)
96
97
98     def gate_verdict(eval_res: dict, floor: Optional[dict]) -> dict:
99         """Pre-flight ship-gate.
100
101         Provisional F1 target (NOT owner-ratified; calibration tracked in #172).
102         Floor (necessary): heuristic LOVO ~0.480 (from quality table).
103         Multi-court ref (makes owned \"worth serving\" per data): 0.66.
104         Proposed for \"very high P/R\" + significance: F1@tIoU0.5 >= 0.70 on
105         multi-court folds with paired-sign p<0.05 vs heuristic (refine as corpus grows).
106         Beating floor + sign test necessary; C3 main commercial blocker.
107         ship=True only when all non-C3 blocks clear (honesty first).
108         """
109         reasons: List[str] = []
110         floor_passed = None
111         sign = None
112         if floor:
113             floor_mean = float(floor["mean_f1"])
114             floor_passed = eval_res["mean_f1"] > floor_mean
115             if not floor_passed:
116                 reasons.append(f"LOVO mean {eval_res['mean_f1']:.3f} <= floor
{floor_mean:.3f}")
117             per = floor.get("per_video", {})
118             paired = [(r["f1"], per[r["video"]]) for r in eval_res["per_video"] if
r["video"] in per]
119             if paired:
120                 wins, n, p = sign_test([lv for lv, _ in paired], [fv for _, fv in paired])
121                 sign = {"wins": wins, "n": n, "p_one_sided": round(p, 4)}
122                 if p >= 0.05:
123                     reasons.append(f"paired sign test not significant ({wins}/{n} wins,
p={p:.3f}) "
124                                   f"? grow the corpus before claiming superiority")
125             else:
126                 reasons.append("no floor baseline provided ? nothing to beat")
127
128         # Provisional blocker (per owner review on #170; see #172 for calibration).
129         # (The number itself is not the blocker; calibration tracked in #172.)
130         reasons.append("ship-gate F1@tIoU target NOT YET CALIBRATED ? provisional; see issue
#172")
131         reasons.append("WASB checkpoint provenance C3 UNRESOLVED ? commercial ship blocked
regardless of F1")
132         # ship remains False until C3 resolved + other gates (by design).
133         return {"ship": False, "floor_passed": floor_passed, "sign_test": sign, "blockers":
reasons}
134
135

```

```

136 def run(manifest_path: str, out_dir: str, version: str,
137         floor_path: Optional[str] = None) -> dict:
138     with open(manifest_path, encoding="utf-8") as f:
139         specs = json.load(f)
140     floor = None
141     if floor_path:
142         with open(floor_path, encoding="utf-8") as f:
143             floor = json.load(f)
144
145     # 1) Honest LOVO eval (the number that matters).
146     eval_res = lovo_run(specs)
147
148     # 2) Final model on all videos.
149     clf, thr = train_final(specs)
150
151     # 3) Gate (pre-flight).
152     verdict = gate_verdict(eval_res, floor)
153
154     # 4) Artifact bundle.
155     bundle_dir = os.path.join(out_dir, version)
156     os.makedirs(bundle_dir, exist_ok=True)
157     weights_path = os.path.join(bundle_dir, "weights.json")
158     payload = clf.to_dict()
159     payload["calibration"] = {"threshold": thr, **CALIBRATION_DEFAULTS}
160     payload["feature_schema_version"] = FEATURE_SCHEMA["version"]
161     with open(weights_path, "w", encoding="utf-8") as f:
162         json.dump(payload, f, indent=2)
163
164     manifest = {
165         "artifact": {
166             "name": "rally_tal_gen0", "version": version,
167             "created_at": time.strftime("%Y-%m-%dT%H:%M:%S%z"),
168             "git_sha": _git_sha(),
169             "weights_file": "weights.json", "weights_sha256": _sha256(weights_path),
170         },
171         "feature_schema": FEATURE_SCHEMA,
172         "calibration": {"threshold": thr, **CALIBRATION_DEFAULTS,
173                        "note": "threshold tuned train-side on the full corpus; "
174                               "honest generalization = the LOVO eval below"},
175         "training_lineage": {
176             "corpus_manifest": manifest_path,
177             "split_protocol": "leave-one-video-out by match",
178             "label_source": "human (rally-annotator golden CSVs)",
179             "videos": [{"name": s["name"], "labels": s["labels"],
180                        "gt_hash": gt_hash(load_gt_intervals(s["labels"],
181strict=False)),
182                        "fps": s["fps"]}] for s in specs],
183         },
184         "eval": {
185             "protocol": "LOVO, F1@tIoU=0.5 segment-level + per-frame AUC",
186             "per_video": eval_res["per_video"],
187             "mean_f1": eval_res["mean_f1"],
188             "floor": floor,
189         },
190         "attestation": {
191             "deps_licenses": "numpy/scipy (BSD) + repo-internal code only (Gen-0)",
192             "wasb_checkpoint_provenance": "C3-UNRESOLVED (ship-blocking; see
193COMMERCIALIZATION C3)",
194             "no_paid_llm_training_labels": True,
195             "consent_ledger": "pending-M0.1 (owner-attested: owned footage, all-adult
196sessions)",
197         },
198         "gate": verdict,
199     }
200     manifest_out = os.path.join(bundle_dir, "manifest.json")

```

```

198     with open(manifest_out, "w", encoding="utf-8") as f:
199         json.dump(manifest, f, indent=2)
200
201     print(f"\n=== Gen-0 artifact bundle: {bundle_dir} ===")
202     print(f"weights : {weights_path}")
203     print(f"manifest: {manifest_out}")
204     print(f"GATE      : ship={verdict['ship']} floor_passed={verdict['floor_passed']} "
205           f"sign_test={verdict['sign_test']}")
206     for b in verdict["blockers"]:
207         print(f" - {b}")
208     return manifest
209
210
211     def main() -> None:
212         ap = argparse.ArgumentParser(description=__doc__)
213         ap.add_argument("--manifest", required=True,
214                         help="corpus manifest JSON:
215 [{name,trajectory,fps,frame_width,labels}, ...]")
216         ap.add_argument("--floor", default=None,
217                         help="floor baseline JSON (e.g.
218 eval_baselines/heuristic_lovo_n6.json)")
219         ap.add_argument("--out", default=os.path.join("artifacts", "rally_tal_gen0"))
220         ap.add_argument("--version", required=True, help="semver for the artifact bundle")
221         args = ap.parse_args()
222         run(args.manifest, args.out, args.version, floor_path=args.floor)
223
224     if __name__ == "__main__":
225         main()

```

5. assistant (2026-06-21T00:40:30.661Z)

The pull didn't fast-forward because my local branch tracks a non-existent remote ref, but the fetch updated `origin/master` (`a555887..e6110ce`). I'll branch explicitly from `origin/master` per convention. Let me explore the worker and tarball-building code.

6. user (2026-06-21T00:40:32.182Z)

```

backend/worker/__init__.py
backend/worker/__main__.py
backend/worker/__pycache__/__init__.cpython-313.pyc
backend/worker/__pycache__/__main__.cpython-313.pyc
training/gen0/harness.py
training/gen0/__init__.py
training/gen0/__pycache__/harness.cpython-313.pyc
training/gen0/__pycache__/__init__.cpython-313.pyc
---WORKER DIR---
total 120
drwxr-xr-x 1 avidu 197609    0 Jun 20 13:48 .
drwxr-xr-x 1 avidu 197609    0 Jun 20 23:06 ..
drwxr-xr-x 1 avidu 197609    0 Jun 20 22:44 __pycache__
drwxr-xr-x 1 avidu 197609    0 May 30 22:04 annotations
drwxr-xr-x 1 avidu 197609    0 Jun 19 13:03 api
drwxr-xr-x 1 avidu 197609    0 Jun 12 15:39 assets
drwxr-xr-x 1 avidu 197609    0 Jun  7 14:14 audio
drwxr-xr-x 1 avidu 197609    0 Jun 20 23:02 config
drwxr-xr-x 1 avidu 197609    0 Jun  7 14:14 detectors
drwxr-xr-x 1 avidu 197609    0 Jun 19 21:38 eval
drwxr-xr-x 1 avidu 197609    0 Jun 11 10:48 features
drwxr-xr-x 1 avidu 197609    0 Jun 19 13:03 infrastructure
-rw-r--r-- 1 avidu 197609 30594 Jun 19 13:03 main.py
drwxr-xr-x 1 avidu 197609    0 Jun 14 16:28 pipeline
drwxr-xr-x 1 avidu 197609    0 Jun 19 14:26 providers
drwxr-xr-x 1 avidu 197609    0 Jun 15 14:11 reporting

```

```
-rw-r--r-- 1 avidu 197609 3009 Jun 18 13:12 results.py
drwxr-xr-x 1 avidu 197609 0 May 30 14:08 sports
drwxr-xr-x 1 avidu 197609 0 Jun 7 14:14 stitcher
drwxr-xr-x 1 avidu 197609 0 Jun 20 15:47 storage
drwxr-xr-x 1 avidu 197609 0 Jun 14 01:32 tools
drwxr-xr-x 1 avidu 197609 0 Jun 20 23:02 utils
drwxr-xr-x 1 avidu 197609 0 Jun 20 14:39 worker
```

7. user (2026-06-21T00:40:33.920Z)

Found 4 files

```
docs\DECOUPLED_COMPUTE\DEPLOY_REPORT_GCP_2026-06-20.md
docs\SETUP_NEW_MACHINE.md
deploy\gcp\README.md
training\gen0\harness.py
```

8. user (2026-06-21T00:40:36.590Z)

```
1      """Worker dispatcher ? pull from storage ? run the pipeline ? push outputs back.
2
3      python -m backend.worker infer --video-uri <uri> --out-uri <prefix> \
4          [--segmenter wasb_hybrid] [--config config.json] [--set k=v ...] [--scratch DIR]
5      python -m backend.worker train ...          # Gen-0 train-from-bucket (follow-up)
6
7      The URI scheme selects the storage backend via ``backend.storage`` (a bare path /
8      ``local://``
9      is an identity passthrough; ``s3://`` / ``gcs://`` is an object store), so existing
10     local runs
11     are byte-for-byte unchanged. Combine with ``RALLY_DETECTOR_IMPL=stub`` and
12     ``--set indexing.skip_ai_handoff=true`` for a **secret-free, GPU-free** run (no Gemini
13     key) ?
14     the path validated end-to-end on DigitalOcean (``docs/SETUP_NEW_MACHINE.md``).
15
16     This is a thin orchestrator: it composes the SAME building blocks the ``backend.main``
17     CLI uses
18     (validator registry ? segmenter ? report), never a forked pipeline.
19     """
20     from __future__ import annotations
21
22     import argparse
23     import csv
24     import json
25     import os
26     import sys
27     import tempfile
28     from typing import Any, List
29
30     from backend import storage
31     from backend.config import load_config
32     from backend.infrastructure.database import Database
33     from backend.pipeline.segmenters.base import segmenter_registry
34     from backend.pipeline.validators.base import registry as validator_registry
35     from backend.reporting import emit_indexer_report
36     from backend.results import Failure
37     from backend.utils.hashing import compute_video_id
38
39
40     def _coerce(v: str) -> Any:
41         """Coerce a ``--set`` string value to bool/int/float where it round-trips, else
42         str."""
43         low = v.lower()
44         if low in ("true", "false"):
45             return low == "true"
46         for cast in (int, float):
47             try:
```

```

43         return cast(v)
44     except ValueError:
45         pass
46     return v
47
48
49     def _apply_overrides(config: Any, sets: List[str]) -> None:
50         """Apply ``a.b.c=value`` overrides onto the typed config (e.g.
indexing.skip_ai_handoff=true)."""
51         for kv in sets or []:
52             key, sep, val = kv.partition("=")
53             if not sep:
54                 raise ValueError(f"--set expects k=v, got {kv!r}")
55             parts = key.split(".")
56             obj = config
57             for p in parts[:-1]:
58                 obj = getattr(obj, p)
59             setattr(obj, parts[-1], _coerce(val))
60
61
62     def _write_intervals_csv(segments: List[dict], path: str) -> None:
63         """Decimal-second ``[start,end)`` rallies + shot count / ending reason ? the join-
friendly
64         artifact (same schema as the rally-annotator labels)."""
65         with open(path, "w", newline="") as f:
66             w = csv.writer(f)
67             w.writerow(["start", "end", "shot_count", "ending_reason"])
68             for s in segments:
69                 w.writerow([
70                     f'{float(s.get("start_time", 0.0)):.3f}',
71                     f'{float(s.get("end_time", 0.0)):.3f}',
72                     s.get("shot_count", ""),
73                     s.get("ending_reason", s.get("shot_count_confidence", "")),
74                 ])
75
76
77     def _write_report_json(video_id: str, segments: List[dict], status: str, path: str) ->
None:
78         with open(path, "w") as f:
79             json.dump({
80                 "video_id": video_id,
81                 "status": status,
82                 "segment_count": len(segments),
83                 "segments": [{"start": float(s.get("start_time", 0.0)),
84                             "end": float(s.get("end_time", 0.0))} for s in segments],
85             }, f, indent=2)
86
87
88     def cmd_infer(args: argparse.Namespace) -> int:
89         config = load_config(args.config) if args.config else load_config()
90         _apply_overrides(config, args.set)
91         storage_cfg = config.storage.model_dump()
92
93         scratch = args.scratch or tempfile.mkdtemp(prefix="rally-worker-")
94         os.makedirs(scratch, exist_ok=True)
95         config.output.output_dir = scratch
96
97         # 1. PULL ? local:// / bare path = identity passthrough; s3:// / gcs:// = download
to scratch.
98         local_video = storage.fetch(args.video_uri, os.path.join(scratch, "in.mp4"),
config=storage_cfg)
99         print(f"[worker] pulled {args.video_uri} -> {local_video}")
100
101         # 2. IDENTITY ? whole-file MD5 (the labels?video join key; pulled bytes must be
byte-identical).

```

```

102     video_id = compute_video_id(local_video)
103
104     # 3. VALIDATE (same registry as the main CLI).
105     val_results, val_context = validator_registry.run_all_with_context(local_video,
106     config)
107     failures = [r for r in val_results if not r.passed]
108     db = Database(os.path.join(scratch, config.output.db_name))
109     db.add_video(video_id, local_video, "failed" if failures else "processed",
110     [r.model_dump() for r in val_results])
111
112     segments: List[dict] = []
113     segmenter_actual = None
114     segmentation_ran = False
115     status = "failed"
116     try:
117         if failures:
118             print("[worker] validation FAILED: "
119             + "; ".join(f"{r.validator_name}: {r.message}" for r in failures))
120             return 2
121         seg_name = args.segmenter or config.indexing.default_segmenter
122         segmenter_cls = segmenter_registry.get(seg_name)
123         if not segmenter_cls:
124             print(f"[worker] unknown segmenter: {seg_name!r} "
125             f"(have {list(segmenter_registry.list_available())})")
126             return 2
127         segmenter_actual = seg_name
128         result = segmenter_cls(db, config).process_video(video_id, local_video,
129         max_frames=args.max_frames)
130         segmentation_ran = True
131         if isinstance(result, Failure):
132             print(f"[worker] segmenter FAILED: {result.message}")
133             return 3
134         segments = result.value if hasattr(result, "value") else result
135         status = "success"
136     finally:
137         # Best-effort per-video observability report (never raises) ? parity with the
138         main CLI.
139         emit_indexer_report(
140             db=db, video_id=video_id, video_path=local_video, config=config,
141             val_results=val_results, val_context=val_context,
142             segmenter_requested=args.segmenter, segmenter_actual=segmenter_actual,
143             was_reingest=False, segmentation_ran=segmentation_ran,
144         )
145
146     # 5. SERIALIZE + 6. PUSH (outputs/<video_id>/?). idempotent on video_id.
147     out_local = os.path.join(scratch, "outputs", video_id)
148     os.makedirs(out_local, exist_ok=True)
149     _write_intervals_csv(segments, os.path.join(out_local, "intervals.csv"))
150     _write_report_json(video_id, segments, status, os.path.join(out_local,
151     "report.json"))
152
153     out_prefix = args.out_uri.rstrip("/")
154     pushed = []
155     for fn in sorted(os.listdir(out_local)):
156         uri = f"{out_prefix}/outputs/{video_id}/{fn}"
157         storage.put(os.path.join(out_local, fn), uri, config=storage_cfg)
158         pushed.append(uri)
159
160     print(f"[worker] infer done: {len(segments)} segment(s); pushed {len(pushed)}
161     artifact(s):")
162     for u in pushed:
163         print("    ", u)
164     return 0

```



```

162     #: Video extensions accepted under <corpus>/videos/ (mirrors the labels/ '.csv' filter
so a
163     #: per-video sidecar ? .srt/.json/thumbnail ? can't shadow the real video on a stem
collision).
164     _VIDEO_EXTS = (".mp4", ".mov", ".mkv", ".avi", ".webm", ".m4v")
165
166
167     def _probe_video_fps_width(path: str):
168         """Robust ffprobe (fps, frame_width) for a real video. Returns None on ANY failure.
169
170         The detector trajectory is frame-indexed; the harness maps frame->time via fps, so a
WRONG
171         fps silently mis-aligns every rally label (e.g. a 60fps clip read as 30 doubles
every time).
172         cv2's ``_probe_fps_width`` silently defaults to (30, 1280) on a read miss ? fine for
the
173         inference report (it flags the fallback) but corrupting for REAL training data. So
here we
174         probe with ffprobe (same tool as ``get_video_duration``) and return None so the
caller SKIPS
175         rather than guesses.
176         """
177         import json
178         import subprocess
179         try:
180             out = subprocess.check_output(
181                 ["ffprobe", "-v", "error", "-select_streams", "v:0",
182                  "-show_entries", "stream=r_frame_rate,width", "-of", "json", path],
183                 stderr=subprocess.DEVNULL).decode()
184             st = (json.loads(out).get("streams") or [{}])[0]
185             num, den = str(st.get("r_frame_rate", "0/1")).partition("/")
186             fps = float(num) / float(den or "1")
187             width = float(st.get("width") or 0)
188             if fps > 0 and width > 0:
189                 return fps, width
190         except (subprocess.SubprocessError, ValueError, KeyError, OSError,
json.JSONDecodeError):
191             pass
192         return None
193
194
195     def _resolve_train_detector(args: argparse.Namespace, config: Any):
196         """Build the trajectory DetectorRunner for ``--trajectory-source detector``, honouring
197         detector_impl (RALLY_DETECTOR_IMPL / indexing.detector_impl: auto=WSL, native=GPU
box,
198         stub=CPU mock). Returns the runner, or prints an error + returns None.
199
200         Reuses the segmenter's ``_resolve_runner`` seam so the worker and the live CLI build
the
201         detector identically (one source of truth). A non-trajectory segmenter (e.g. yolo)
has
202         no shuttle detector and is rejected.
203         """
204         from backend.pipeline.segmenters.trajectory_hybrid import TrajectoryHybridSegmenter
205
206         seg_cls = segmenter_registry.get(args.segmenter)
207         if not seg_cls:
208             print(f"[worker] unknown segmenter: {args.segmenter!r} "
209                   f"(have {list(segmenter_registry.list_available())})")
210             return None
211         seg = seg_cls(None, config)
212         if not isinstance(seg, TrajectoryHybridSegmenter):
213             print(f"[worker] --segmenter {args.segmenter!r} has no shuttle detector; "
214                   f"use a trajectory hybrid (wasb_hybrid / tracknet_hybrid /
fusion_hybrid).")

```

```

215         return None
216     try:
217         runner, _ = seg._resolve_runner(config.get("indexing", {}))
218     except NotImplementedError as e:
219         # e.g. detector_impl='native' for a family without a native runner (tracknet, or
220         # fusion with a tracknet substrate). Fail cleanly (rc!=0), not with a raw
221         traceback.
222         print(f"[worker] detector not available: {e}")
223         return None
224     ok, msg = runner.healthcheck()
225     if not ok:
226         print(f"[worker] detector environment not ready: {msg}")
227         return None
228     print(f"[worker] detector ready ({args.segmenter}): {msg}")
229     return runner
230
231 def cmd_train(args: argparse.Namespace) -> int:
232     """Gen-0 train-from-bucket: pull labels (+ videos) -> trajectories -> manifest ->
233     shell out
234     to training.gen0.harness (backend must NOT import training) -> push the artifact
235     bundle.
236     Two trajectory sources (the train pipeline + harness are identical either way):
237     * ``synth`` (default) ? the CPU "mock GPU": deterministic, label-consistent
238     synthetic
239     shuttle paths (no GPU/WSL), so the whole pipeline is validated on a CPU box /
240     CI.
241     * ``detector`` ? REAL shuttle trajectories: pull each video and run the resolved
242     DetectorRunner (native WASB on a GPU box, or stub for a GPU-free wiring test).
243     This
244     is the M3.5 "first real training" path; only the trajectory source flips.
245     """
246     import subprocess
247
248     config = load_config(args.config) if args.config else load_config()
249     _apply_overrides(config, args.set)
250     storage_cfg = config.storage.model_dump()
251
252     scratch = args.scratch or tempfile.mkdtemp(prefix="rally-train-")
253     labels_dir = os.path.join(scratch, "labels")
254     traj_dir = os.path.join(scratch, "trajectories")
255     videos_dir = os.path.join(scratch, "videos")
256     os.makedirs(labels_dir, exist_ok=True)
257     os.makedirs(traj_dir, exist_ok=True)
258
259     corpus = args.corpus_uri.rstrip("/")
260     label_uris = sorted(u for u in storage.list_uris(f"{corpus}/labels/",
261     config=storage_cfg)
262         if u.lower().endswith(".csv"))
263     if len(label_uris) < 2:
264         print(f"[worker] need >=2 labeled videos for leave-one-video-out; found
265     {len(label_uris)} "
266         f"under {corpus}/labels/")
267     return 2
268
269 # Real-detector source: resolve the runner once + index the bucket's videos/ by
270 stem.
271 runner = None
272 video_by_stem: dict = {}
273 if args.trajectory_source == "detector":
274     os.makedirs(videos_dir, exist_ok=True)
275     runner = _resolve_train_detector(args, config)
276     if runner is None:
277         return 2

```

```

271         for vuri in storage.list_uris(f"{corpus}/videos/", config=storage_cfg):
272             vname = storage.parse_uri(vuri).name
273             if not vname.lower().endswith(_VIDEO_EXTS):
274                 continue # skip sidecars (.srt/.json/thumbnails) so they can't shadow
the video
275                 video_by_stem[os.path.splitext(vname)[0]] = vuri
276
277         print(f"[worker] trajectory source: {args.trajectory_source}")
278         specs: List[dict] = []
279         for uri in label_uris:
280             fname = storage.parse_uri(uri).name # e.g.
GX020094_proxy.rallies.csv
281             name = fname.replace(".rallies.csv", "").replace(".csv", "")
282             local_label = storage.fetch(uri, os.path.join(labels_dir, fname),
config=storage_cfg)
283             if args.trajectory_source == "detector":
284                 vuri = video_by_stem.get(name)
285                 if not vuri:
286                     print(f"[worker] no video for {name!r} under {corpus}/videos/ ?
skipping.")
287                     continue
288                 local_video = storage.fetch(vuri, os.path.join(videos_dir,
storage.parse_uri(vuri).name),
289                                     config=storage_cfg)
290                 video_id = compute_video_id(local_video)
291                 # Real fps/width drive the trajectory frame->time mapping; PROBE (don't
guess) ? a
292                 # wrong fps silently mis-aligns every label, so skip the video if ffprobe
can't read it.
293                 meta = _probe_video_fps_width(local_video)
294                 if meta is None:
295                     print(f"[worker] could not probe fps/width for {name!r} (ffprobe) ?
skipping (won't guess).")
296                     continue
297                 fps, frame_width = meta
298                 traj = runner.run_predict(local_video, traj_dir, video_id=video_id)
299                 if not traj:
300                     print(f"[worker] detector produced no trajectory for {name!r} ?
skipping.")
301                     continue
302                 specs.append({"name": name, "trajectory": traj, "labels": local_label,
303                             "fps": fps, "frame_width": frame_width})
304             else:
305                 from backend.pipeline.detectors.stub_runner import
synth_trajectory_from_labels
306                 traj = os.path.join(traj_dir, f"{name}_traj.csv")
307                 synth_trajectory_from_labels(local_label, traj, fps=args.fps,
frame_width=args.frame_width)
308                 specs.append({"name": name, "trajectory": traj, "labels": local_label,
309                             "fps": args.fps, "frame_width": args.frame_width})
310
311         if len(specs) < 2:
312             print(f"[worker] need >=2 videos with trajectories for leave-one-video-out; got
{len(specs)}.")
313             return 2
314         manifest_path = os.path.join(scratch, "manifest.json")
315         with open(manifest_path, "w", encoding="utf-8") as f:
316             json.dump(specs, f, indent=2)
317         src_label = "real detector" if args.trajectory_source == "detector" else "CPU-mock
stub"
318         print(f"[worker] built manifest: {len(specs)} videos ({src_label} trajectories)")
319
320         out_dir = os.path.join(scratch, "artifacts")
321         cmd = [sys.executable, "-m", "training.gen0.harness",
322               "--manifest", manifest_path, "--out", out_dir, "--version", args.version]

```

```

323         if args.floor:
324             floor_local = (storage.fetch(args.floor, os.path.join(scratch, "floor.json"),
config=storage_cfg)
325                             if "://" in args.floor else args.floor)
326             cmd += ["--floor", floor_local]
327         print("[worker] training:", " ".join(cmd))
328         rc = subprocess.run(cmd).returncode
329         if rc != 0:
330             print(f"[worker] gen0 harness failed (rc={rc})")
331             return rc
332
333         # PUSH the versioned artifact bundle (weights.json + manifest.json) to the bucket.
334         bundle = os.path.join(out_dir, args.version)
335         out_prefix = args.out_uri.rstrip("/")
336         pushed = []
337         for fn in sorted(os.listdir(bundle)):
338             uri = f"{out_prefix}/weights/{args.version}/{fn}"
339             storage.put(os.path.join(bundle, fn), uri, config=storage_cfg)
340             pushed.append(uri)
341         print(f"[worker] train done: pushed {len(pushed)} artifact(s):")
342         for u in pushed:
343             print("    ", u)
344         return 0
345
346
347     def build_parser() -> argparse.ArgumentParser:
348         p = argparse.ArgumentParser(prog="python -m backend.worker",
349                                     description="Storage-aware worker: pull ? run pipeline ?
push.")
350         sub = p.add_subparsers(dest="cmd", required=True)
351
352         pi = sub.add_parser("infer", help="run inference on one video URI and push outputs")
353         pi.add_argument("--video-uri", required=True, help="local path / local:// / s3:// /
gcs://")
354         pi.add_argument("--out-uri", required=True, help="output prefix
(outputs/<video_id>/? is appended)")
355         pi.add_argument("--segmenter", default=None, help="default:
indexing.default_segmenter")
356         pi.add_argument("--config", default=None, help="config.json path (default:
./config.json)")
357         pi.add_argument("--scratch", default=None, help="local scratch dir (default: a temp
dir)")
358         pi.add_argument("--max-frames", type=int, default=None)
359         pi.add_argument("--set", action="append", default=[], metavar="k=v",
360                         help="config override, e.g. --set indexing.skip_ai_handoff=true")
361         pi.set_defaults(func=cmd_infer)
362
363         pt = sub.add_parser("train", help="Gen-0 train-from-bucket (labels [+videos] ->
trajectories -> harness)")
364         pt.add_argument("--corpus-uri", required=True, help="prefix containing labels/ (+
videos/ for "
365                        "--trajectory-source detector), e.g. s3://bucket or gcs://bucket")
366         pt.add_argument("--out-uri", required=True, help="output prefix (weights/<version>/
is appended)")
367         pt.add_argument("--version", required=True, help="semver for the artifact bundle")
368         pt.add_argument("--trajectory-source", choices=["synth", "detector"],
369                         default="synth",
370                         help="synth = CPU-mock label-consistent trajectories (default, no
GPU); "
371                         "detector = REAL shuttle trajectories via the resolved
DetectorRunner "
372                         "(needs videos/ in the corpus; honours detector_impl).")
373         pt.add_argument("--segmenter", default="wasb_hybrid",
374                         help="trajectory hybrid whose detector is used for --trajectory-
source detector")

```

```

374         pt.add_argument("--floor", default=None, help="floor baseline JSON (local path or
storage URI)")
375         pt.add_argument("--config", default=None)
376         pt.add_argument("--scratch", default=None)
377         pt.add_argument("--fps", type=float, default=30.0, help="synth trajectory fps
(detector probes the video)")
378         pt.add_argument("--frame-width", type=float, default=1280.0,
379                         help="synth trajectory frame width (detector probes the video)")
380         pt.add_argument("--set", action="append", default=[], metavar="k=v")
381         pt.set_defaults(func=cmd_train)
382         return p
383
384
385     def main(argv=None) -> int:
386         args = build_parser().parse_args(argv)
387         return args.func(args)
388
389
390     if __name__ == "__main__":
391         sys.exit(main())
392

```

9. user (2026-06-21T00:40:38.269Z)

```

deploy\gcp\README.md:153:git archive --format=tar.gz -o /tmp/rally.tgz HEAD
deploy\gcp\README.md:155:gcloud compute ssh rally-gpu --zone=asia-south1-a -- 'mkdir -p ~/rally
&& tar -xzf ~/rally.tgz -C ~/rally'

```

10. assistant (2026-06-21T00:40:41.824Z)

Now let me look at the deploy scripts that build the tarball and any existing harness tests.

11. user (2026-06-21T00:40:42.969Z)

```

total 24
drwxr-xr-x 1 avidu 197609    0 Jun 20 23:05 .
drwxr-xr-x 1 avidu 197609    0 Jun 20 14:51 ..
-rw-r--r-- 1 avidu 197609 15089 Jun 20 23:05 README.md
-rwxr-xr-x 1 avidu 197609  3114 Jun 20 14:51 provision.sh
-rwxr-xr-x 1 avidu 197609  2788 Jun 20 22:51 teardown.sh
---
total 24
drwxr-xr-x 1 avidu 197609    0 Jun 20 23:04 .
drwxr-xr-x 1 avidu 197609    0 Jun 20 14:51 ..
-rw-r--r-- 1 avidu 197609  4844 Jun 20 23:04 README.md
-rwxr-xr-x 1 avidu 197609  2433 Jun 20 13:48 bootstrap.sh
-rwxr-xr-x 1 avidu 197609  3537 Jun 20 20:31 gpu_setup.sh
-rwxr-xr-x 1 avidu 197609  1939 Jun 20 13:48 mount_scratch.sh

```

12. user (2026-06-21T00:40:44.300Z)

```

deploy\digitalocean\gpu_setup.sh:52:WEIGHTS="$WASB_REPO/pretrained_weights/wasb_badminton_best.p
th.tar"
deploy\digitalocean\gpu_setup.sh:56:  echo "[gpu]          !! MISSING $WEIGHTS ? fetch
wasb_badminton_best.pth.tar (Drive/bucket) before extraction."
deploy\gcp\teardown.sh:27:  echo "== stopping $VM ? ? the DISK STILL BILLS (~$/mo); use
'delete' for \$0. Resume: gcloud compute instances start $VM --zone=$ZONE =="

```

13. user (2026-06-21T00:40:44.844Z)

```

tests\test_eval_harness.py
.claude\worktrees\bhi-execpolicy\tests\test_eval_harness.py
.claude\worktrees\agent-a771ff1d85e18f2ed\tests\test_eval_harness.py
.claude\worktrees\tier1-windowing\tests\test_eval_harness.py

```

```
.claude\worktrees\w2-safeguard\tests\test_eval_harness.py
.claude\worktrees\hardchop\tests\test_eval_harness.py
.claude\worktrees\boxhill17\tests\test_eval_harness.py
.claude\worktrees\pr-stitch\tests\test_eval_harness.py
.claude\worktrees\gap-doc\tests\test_eval_harness.py
.claude\worktrees\resweep\tests\test_eval_harness.py
.claude\worktrees\recdoc\tests\test_eval_harness.py
```

14. user (2026-06-21T00:40:45.290Z)

```
.claude\worktrees\bhi-execpolicy\tests\test_training_gen0.py
.claude\worktrees\bhi-
execpolicy\tests\__pycache__\test_training_gen0.cpython-313-pytest-9.0.3.pyc
tests\test_training_gen0.py
.claude\worktrees\agent-a771ff1d85e18f2ed\tests\test_training_gen0.py
.claude\worktrees\agent-a771ff1d85e18f2ed\tests\__pycache__\test_training_gen0.cpython-313-
pytest-9.0.3.pyc
tests\__pycache__\test_training_gen0.cpython-313-pytest-9.0.3.pyc
.claude\worktrees\tier1-windowing\tests\test_training_gen0.py
.claude\worktrees\tier1-windowing\tests\__pycache__\test_training_gen0.cpython-313-pytest-
9.0.3.pyc
.claude\worktrees\w2-safeguard\tests\test_training_gen0.py
.claude\worktrees\w2-safeguard\tests\__pycache__\test_training_gen0.cpython-313-pytest-9.0.3.pyc
.claude\worktrees\hardchop\tests\test_training_gen0.py
.claude\worktrees\hardchop\tests\__pycache__\test_training_gen0.cpython-313-pytest-9.0.3.pyc
.claude\worktrees\boxhill17\tests\test_training_gen0.py
.claude\worktrees\pr-stitch\tests\test_training_gen0.py
.claude\worktrees\pr-stitch\tests\__pycache__\test_training_gen0.cpython-313-pytest-9.0.3.pyc
.claude\worktrees\gap-doc\tests\test_training_gen0.py
.claude\worktrees\resweep\tests\test_training_gen0.py
.claude\worktrees\recdoc\tests\test_training_gen0.py
```

15. assistant (2026-06-21T00:40:48.560Z)

Let me read the existing test file and the deploy README's tarball section.

16. user (2026-06-21T00:40:50.001Z)

```
1      """Gen-0 harness (training/gen0): end-to-end on SYNTHETIC fixtures only.
2
3      Real trajectories/labels are never committed (training/README.md rule 4 ? consent ledger
4      pending). The synthetic corpus mirrors tests/test_rally_seq_proto.py: separable rally vs
5      dead-time signal, three rallies per 'video'.
6      """
7      import csv
8      import json
9
10     import numpy as np
11     import pytest
12
13     from training.gen0 import harness
14
15
16     def _write_traj(path, fps=10.0):
17         rows = [("Frame", "Visibility", "X", "Y")]
18         n = int(120 * fps)
19         for f in range(n):
20             t = f / fps
21             in_rally = (t < 20) or (40 <= t < 60) or (80 <= t < 100)
22             if in_rally:
23                 x = 50 + 45 * np.sin(2 * np.pi * (f % 10) / 10.0)
24                 rows.append((f, 1, round(float(x), 2), 50))
25             else:
26                 rows.append((f, 1, 50.0, 50) if f % 5 == 0 else (f, 0, -1, -1))
```

```

27         with open(path, "w", newline="") as fh:
28             csv.writer(fh).writerows(rows)
29         return str(path)
30
31
32     def _write_labels(path):
33         with open(path, "w", newline="", encoding="utf-8") as fh:
34             w = csv.writer(fh)
35             w.writerow(["rally_number", "start_time", "end_time"])
36             w.writerow([1, 0.0, 20.0])
37             w.writerow([2, 40.0, 60.0])
38             w.writerow([3, 80.0, 100.0])
39         return str(path)
40
41
42     @pytest.fixture
43     def corpus(tmp_path):
44         specs = [{
45             "name": f"v{i}", "trajectory": _write_traj(tmp_path / f"t{i}.csv"),
46             "fps": 10.0, "frame_width": 100.0, "labels": _write_labels(tmp_path /
47 f"g{i}.csv"),
48         } for i in range(2)]
49         mpath = tmp_path / "manifest.json"
50         mpath.write_text(json.dumps(specs), encoding="utf-8")
51         return str(mpath), tmp_path
52
53     def test_harness_end_to_end_emits_artifact_bundle(corpus):
54         mpath, tmp_path = corpus
55         floor = tmp_path / "floor.json"
56         floor.write_text(json.dumps({
57             "name": "synthetic-floor", "mean_f1": 0.10,
58             "per_video": {"v0": 0.10, "v1": 0.05},
59         }), encoding="utf-8")
60
61         out = harness.run(mpath, str(tmp_path / "artifacts"), "0.0.1",
62 floor_path=str(floor))
63
64         bundle = tmp_path / "artifacts" / "0.0.1"
65         assert (bundle / "weights.json").exists()
66         assert (bundle / "manifest.json").exists()
67
68         # Manifest contract (the reviewable surface).
69         m = json.loads((bundle / "manifest.json").read_text(encoding="utf-8"))
70         assert m["artifact"]["name"] == "rally_tal_gen0"
71         assert m["artifact"]["weights_sha256"]
72         assert m["feature_schema"]["version"] >= 1
73         assert m["feature_schema"]["fps_semantics"] == "per-second"
74         assert 0.2 <= m["calibration"]["threshold"] <= 0.8
75         assert len(m["training_lineage"]["videos"]) == 2
76         assert all(v["gt_hash"] for v in m["training_lineage"]["videos"])
77         assert m["eval"]["mean_f1"] > 0.5 # separable synthetic signal
78         assert m["attestation"]["no_paid_llm_training_labels"] is True
79
80         # Gate: floor beaten, but NEVER self-ships (provisional target + C3 open; see #172).
81         g = m["gate"]
82         assert g["ship"] is False
83         assert g["floor_passed"] is True
84         assert any("C3" in b for b in g["blockers"])
85         assert any("NOT YET CALIBRATED" in b for b in g["blockers"])
86         assert any("#172" in b for b in g["blockers"])
87         assert out["gate"] == g
88
89     def test_weights_round_trip_and_schema_pin(corpus):

```

```

90     mpath, tmp_path = corpus
91     harness.run(mpath, str(tmp_path / "artifacts"), "0.0.2", floor_path=None)
92     payload = json.loads((tmp_path / "artifacts" / "0.0.2" /
"weights.json").read_text())
93     from backend.eval.classifier import LogisticRegression
94     from backend.features.rally_seq import FEAT_NAMES, FEATURE_SCHEMA_VERSION
95     clf = LogisticRegression.from_dict(payload)
96     assert payload["feature_schema_version"] == FEATURE_SCHEMA_VERSION
97     assert list(clf.feature_names) == FEAT_NAMES
98     probs = clf.predict_proba(np.zeros((4, len(FEAT_NAMES))))
99     assert probs.shape == (4,)
100
101
102     def test_sign_test_math():
103         # 5/6 wins -> one-sided p = 7/64 ~ 0.109 (NOT significant ? honesty pinned).
104         wins, n, p = harness.sign_test([1, 1, 1, 1, 1, 0], [0, 0, 0, 0, 0, 1])
105         assert (wins, n) == (5, 6)
106         assert p == pytest.approx(7 / 64)
107         # 6/6 -> p = 1/64 < 0.05.
108         _, _, p6 = harness.sign_test([1] * 6, [0] * 6)
109         assert p6 == pytest.approx(1 / 64)
110

```

17. user (2026-06-21T00:40:50.571Z)

```

130     > `--metadata=startup-script` so observability is automatic on every box.
131
132     > **Long remote steps must be detached.** A plain `ssh ? "<long cmd>"` dies if the SSH
drops (seen
133     > mid-build, 2026-06-20) ? run them `nohup`'d on the box (`nohup bash setup.sh >log 2>&1
&`) and poll
134     > the log, so a dropped connection can't kill the install.
135
136     > If `asia-south1-a` reports no T4 capacity, try `-b` / `-c`. The Ubuntu image needs the
NVIDIA
137     > driver: either add `--metadata=install-nvidia-driver=True` style startup, or after SSH
run GCP's
138     > driver installer, **or** use a Deep-Learning-VM image (`--image-family=common-
cu124-ubuntu-2204`
139     > `--image-project=deeplearning-platform-release`) which ships the driver. Confirm
`nvidia-smi` works
140     > before $4.
141
142     SSH: `gcloud compute ssh rally-gpu --zone=asia-south1-a`.
143
144     ---
145
146     ## 4. Port the repo + WASB env onto the box
147
148     The `deploy/digitalocean/` scripts are **provider-agnostic** (they detect the local NVMe
/ install
149     torch); reuse them on GCP:
150
151     ```bash
152     # from your machine ? ship the repo:
153     git archive --format=tar.gz -o /tmp/rally.tgz HEAD
154     gcloud compute scp /tmp/rally.tgz rally-gpu:/root/ --zone=asia-south1-a
155     gcloud compute ssh rally-gpu --zone=asia-south1-a -- 'mkdir -p ~/rally && tar -xzf
~/rally.tgz -C ~/rally'
156
157     # on the box:
158     cd ~/rally
159     sudo bash deploy/digitalocean/mount_scratch.sh && export TMPDIR=/scratch # detects the
local NVMe
160     ./deploy/digitalocean/bootstrap.sh --gpu # venv +

```



```

torch cu124 (confirm cuda True)
161     . .venv/bin/activate && pip install -e .[storage-gcs]           # GCS backend
162     bash deploy/digitalocean/gpu_setup.sh                          # native
`wasb` conda env + WASB-SBDT
163
164     # secrets (NOT in repo): the GCS SA JSON + a rotated Gemini key
165     gcloud compute scp ~/.gcp/rally-worker-sa.json rally-gpu:/root/.gcp/sa.json --zone=asia-
south1-a
166     # on the box: export GOOGLE_APPLICATION_CREDENTIALS=/root/.gcp/sa.json
167     mkdir -p /root/.keys && printf '%s' '<ROTATED_GEMINI_KEY>' > /root/.keys/GEMINI_API_KEY
&& chmod 600 /root/.keys/GEMINI_API_KEY
168     ```
169
170     WASB weights (`wasb_badminton_best.pth.tar`) must be present at
171     `~/models/WASB-SBDT/pretrained_weights/` (gpu_setup.sh does NOT fetch them ? stage from
Drive/bucket).
172     Point the native runner at them: `export WASB_WEIGHTS=~/models/WASB-
SBDT/pretrained_weights/wasb_badminton_best.pth.tar`
173     and `export WASB_PYTHON=$(conda run -n wasb which python)` (or the env's python path).
See
174     `docs/TRACKNET_WSL_SETUP.md` for the WASB env. **WASB-C3:** weight provenance is an open
legal item
175     before sharing any trajectories/model externally ? internal extraction + Gen-0 is fine.
176
177     ---
178
179     ## 5. Stage the corpus + run the first REAL train/infer

```

18. assistant (2026-06-21T00:42:36.740Z)

I now have the full picture. Let me check `.gitignore` and confirm the repo-root layout before implementing.

19. user (2026-06-21T00:42:38.950Z)

```

=== repo root files ===
pyproject.toml
requirements.txt
ruff.toml
run_all_tests.py
run_eval.py
run_regression.py
=== .gitignore (git_sha/GIT_SHA refs) ===
=== git 2.38 add-file check ===
git version 2.45.2.windows.1

```

20. user (2026-06-21T00:42:39.847Z)

```

tests\test_ablation.py:122:def test_unknown_gate_knob_raises():
tests\test_annotation_providers.py:23:def test_registry_unknown_provider_raises():
training\gen0\harness.py:50:def _git_sha() -> str:
training\gen0\harness.py:53:                                     text=True, timeout=10).stdout.strip()
or "unknown"
training\gen0\harness.py:55:         return "unknown"
training\gen0\harness.py:168:         "git_sha": _git_sha(),
tests\test_async_jobs.py:211:def test_submit_unknown_segmenter_400_and_no_job_row(env):
tests\test_async_jobs.py:307:def test_get_unknown_job_is_404(env):
tests\test_api_endpoints.py:230:     s_unknown = db.add_play_segment("vid", 20.0, 24.0) # no
shot_count metadata
tests\test_api_endpoints.py:244:         "video_id": "vid", "segment_ids": [s_low, s_high,
s_unknown],
tests\test_automated_provider.py:44:def test_poll_unknown_job_is_failed():
tests\test_automated_provider.py:49:def test_fetch_unknown_job_raises():
tests\test_config.py:155:def test_unknown_keys_warn_with_dotted_paths(tmp_path, caplog):

```

```

tests\test_config.py:156:      """Typo'd keys must be LOUD: top-level extras are kept-but-unread
and unknown
tests\test_config.py:163:      "validation": {"relevance": {"unknown_gate": True}}, # doubly
nested
tests\test_config.py:172:      assert "validation.relevance.unknown_gate" in joined
tests\test_config.py:262: def test_config_unknown_key_paths_warning_and_tolerate():
tests\test_config.py:263:      """Positive test for Item 2: _unknown_key_paths warning for unknown
top-level keys,
tests\test_config.py:264:      but tolerated (extra='allow'), and unknown section keys dropped
silently.
tests\test_config.py:267:      p = os.path.join(td, "unknown.json")
tests\test_config.py:271:      "completely_unknown_top": "bar",
tests\test_config.py:272:      "validation": {"unknown_in_section": 99} # dropped
tests\test_config_wasb.py:4: nested sections silently dropped unknown keys and the whole
indexing.wasb block (the live
tests\test_config_wasb.py:54:      # Must be DECLARED (nested sections drop unknown keys) and
default to 1280 ?
backend\worker\_main__.py:123:      print(f"[worker] unknown segmenter: {seg_name!r} ")
backend\worker\_main__.py:208:      print(f"[worker] unknown segmenter: {args.segmenter!r} ")
tests\test_eval_harness.py:41: def test_get_predictions_unknown_stage_raises(db):
backend\annotations\base.py:93:      key = getattr(provider_cls, "__name__", "unknown")
tests\test_execution_policy.py:48: def test_policy_roundtrip_and_unknown_keys_ignored():
backend\eval\ablation.py:222:      raise ValueError(f"unknown gate_knob {gate_knob!r}
(expected 'min_crossings'/'min_players'"))
backend\annotations\automated_provider.py:99:      raise KeyError(f"AutomatedProvider:
unknown job '{job_id}' (submit it first)")
tests\test_experiment.py:119: def test_load_golden_videos_rejects_unknown_audio_policy(tmp_path):
tests\test_experiment.py:137: def test_from_dict_ignores_unknown_keys():
backend\eval\experiment.py:345:      raise ExperimentError(f"unknown
calibrate={variant.calibrate!r} (use 'platt'/'isotonic'"))
backend\eval\golden_fixtures.py:198:      raise ValueError(f"unknown gap kind
{spec.kind!r} (use 'still'/'occluded'"))
backend\eval\golden_fixtures.py:211:      raise TypeError(f"unknown segment type
{type(seg).__name__}")
backend\config\loader.py:25: def _unknown_key_paths(data: Dict[str, Any], model_cls:
type[BaseModel], prefix: str = "") -> list[str]:
backend\config\loader.py:33:      unknown: list[str] = []
backend\config\loader.py:37:      unknown.append(prefix + key)
backend\config\loader.py:41:      unknown.extend(_unknown_key_paths(val, ann,
f"{prefix}{key}.")
backend\config\loader.py:42:      return unknown
backend\config\loader.py:76:      # otherwise crash startup: a truthy non-mapping hits `.items()`
in _unknown_key_paths, and any
backend\config\loader.py:85:      unknown = _unknown_key_paths(file_data, AppConfig)
backend\config\loader.py:86:      if unknown:
backend\config\loader.py:90:          "but unread; unknown section keys are silently
dropped): %s",
backend\config\loader.py:91:          cfg_path, ", ".join(sorted(unknown)),
backend\config\models.py:378:      # Re-raise like a real dict for unknown top-level
backend\eval\harness.py:62:      raise ValueError(f"unknown stage {stage!r}; expected one of
{STAGES}")
backend\eval\rally_seg_eval.py:412:      raise SystemExit(f"unknown preset '{name}'; choices:
{sorted(PRESETS)}")
backend\utils\geometry.py:39:      return raw_velocity # Fallback to raw pixels if width
unknown
backend\main.py:174:      # Submit-time validation already 400s unknown names; this is the
defensive
backend\main.py:244:      Fast-fail checks (bad path, missing file, unknown segmenter) happen here
so the
backend\main.py:375:      pass # unparseable count -> treat as unknown, keep
backend\reporting\harvest.py:34:      meta: Dict[str, Any] = {"fps_source": "unknown"}
backend\reporting\harvest.py:209:      if vm.get("fps_source") == "unknown" and segmentation_ran:
backend\storage\_init__.py:41:      raise ValueError(f"unknown storage backend: {name!r}
(expected local|s3|gcs|gdrive)")
backend\pipeline\stitcher\compiler.py:333:      logger.warning("Video duration unknown.

```

```

Cannot validate segment boundaries against video length.")
backend\pipeline\segmenters\gemini.py:70:      early-return cases (unknown sport, missing API
key, unknown provider) ? the
backend\pipeline\segmenters\gemini.py:508:      # Resolve sport profile + provider (handles
the unknown-sport / missing-key / unknown-provider early returns)
backend\pipeline\segmenters\yolo_hybrid.py:62:      or unknown provider). Behaviour is
identical to the inlined version.
backend\pipeline\detectors\wasb_infer.py:593:      """Container-reported frame count via cv2 (0 if
unknown). Used to size the stream."""
tests\test_rally_seg_eval.py:153:      assert court_type_for({"name": "unknown_clip"}) == "single"
tests\test_rally_seg_eval_report.py:300:def test_main_unknown_preset_exits(monkeypatch,
tmp_path):
tests\test_rally_slicer.py:64:def test_subset_and_unknown_skipped():
tests\test_rally_slicer.py:66:      w = compute_clip_windows(labels, ["3", "1", "99"]) # 99
unknown -> dropped
tests\test_storage.py:39:def test_parse_uri_rejects_unknown_scheme():
tests\test_storage.py:95:def test_get_storage_unknown_backend_raises():
tests\test_upload_download.py:112:def test_unknown_upload_id_is_404(env):
tests\test_upload_download.py:138:      # aborted: id unknown afterwards, no partials on disk
tests\test_upload_download.py:164:def test_download_unknown_video_404(env):

```

21. assistant (2026-06-21T00:42:43.341Z)

Everything's clear. Let me create the branch off `origin/master` (explicit, per the branching-safety convention) and verify HEAD.

22. user (2026-06-21T00:42:46.744Z)

```

branch 'fix/gen0-git-sha-provenance' set up to track 'origin/master'.
Switched to a new branch 'fix/gen0-git-sha-provenance'
--- HEAD ---
e6110ce feat(api): GET /api/health + return download_url from /api/compile (M1/P6) (#268)
--- branch ---
fix/gen0-git-sha-provenance

```

23. assistant (2026-06-21T00:42:52.410Z)

On a clean branch from `origin/master` at `e6110ce`. Now let me implement the fix in the harness.

24. user (2026-06-21T00:42:54.173Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\training\gen0\harness.py has been updated successfully. (file state is current in your context ? no need to Read it back)

25. user (2026-06-21T00:43:07.400Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\training\gen0\harness.py has been updated successfully. (file state is current in your context ? no need to Read it back)

26. assistant (2026-06-21T00:43:13.436Z)

Now let me add the unit tests covering the resolution chain and manifest stamping.

27. user (2026-06-21T00:43:14.873Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_training_gen0.py has been updated successfully. (file state is current in your context ? no need to Read it back)

28. user (2026-06-21T00:43:29.293Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_training_gen0.py has been updated successfully. (file state is current in your context ? no need to Read it back)

29. assistant (2026-06-21T00:43:33.204Z)

Now let me update the deploy README so the tarball-build flow actually stamps the SHA (otherwise the fix stays latent).

30. user (2026-06-21T00:43:33.725Z)

```
146     ## 4. Port the repo + WASB env onto the box
147
148     The `deploy/digitalocean/` scripts are **provider-agnostic** (they detect the local NVMe
/ install
149     torch); reuse them on GCP:
150
151     ```bash
152     # from your machine ? ship the repo:
153     git archive --format=tar.gz -o /tmp/rally.tgz HEAD
154     gcloud compute scp /tmp/rally.tgz rally-gpu:/root/ --zone=asia-south1-a
155     gcloud compute ssh rally-gpu --zone=asia-south1-a -- 'mkdir -p ~/rally && tar -xzf
~/rally.tgz -C ~/rally'
156
157     # on the box:
158     cd ~/rally
159     sudo bash deploy/digitalocean/mount_scratch.sh && export TMPDIR=/scratch # detects the
local NVMe
160     ./deploy/digitalocean/bootstrap.sh --gpu # venv +
torch cu124 (confirm cuda True)
161     . .venv/bin/activate && pip install -e .[storage-gcs] # GCS backend
162     bash deploy/digitalocean/gpu_setup.sh # native
`wasb` conda env + WASB-SBDT
163
164     # secrets (NOT in repo): the GCS SA JSON + a rotated Gemini key
165     gcloud compute scp ~/.gcp/rally-worker-sa.json rally-gpu:/root/.gcp/sa.json --zone=asia-
south1-a
```

31. user (2026-06-21T00:43:42.998Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\deploy\gcp\README.md has been updated successfully. (file state is current in your context ? no need to Read it back)

32. assistant (2026-06-21T00:43:46.626Z)

Let me add `GIT\_SHA` to `.gitignore` as a safety so a locally-generated one can't be committed.

33. user (2026-06-21T00:43:47.503Z)

```
1     # Byte-compiled / optimized / DLL files
2     __pycache__/
3     *.py[cod]
4     *$py.class
5     *.pt
6
7     # Local environment secrets
8     .env
9     *.env
10    .env.*
11
12    # C extensions
13    *.so
14
15    # Distribution / packaging
16    .Python
17    build/
18    develop-eggs/
19    dist/
```

```
20     downloads/
21     eggs/
22     .eggs/
23     lib/
24     lib64/
25     parts/
26     sdist/
27     var/
28     wheels/
29     share/python-wheels/
30     *.egg-info/
```

34. user (2026-06-21T00:43:52.014Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\.gitignore has been updated successfully. (file state is current in your context ? no need to Read it back)

35. assistant (2026-06-21T00:43:55.254Z)

Now let me run the new tests, then lint/type-check the changed files.

36. user (2026-06-21T00:44:01.392Z)

```
..... [100%]
9 passed in 4.09s
```

37. assistant (2026-06-21T00:44:04.302Z)

All 9 pass. Now lint + type-check the changed Python files.

38. user (2026-06-21T00:44:06.397Z)

```
=== ruff ===
All checks passed!
=== mypy ===
backend\annotations\base.py:85: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs [annotation-unchecked]
training\gen0\harness.py:85: error: Incompatible return value type (got "str | None", expected
"str") [return-value]
Found 1 error in 1 file (checked 1 source file)
```

39. assistant (2026-06-21T00:44:13.958Z)

mypy can't narrow the double-call. Let me assign once.

40. user (2026-06-21T00:44:16.137Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\training\gen0\harness.py has been updated successfully. (file state is current in your context ? no need to Read it back)

41. user (2026-06-21T00:44:20.335Z)

```
=== mypy ===
backend\annotations\base.py:85: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs [annotation-unchecked]
Success: no issues found in 1 source file
=== ruff ===
All checks passed!
```

42. assistant (2026-06-21T00:44:25.704Z)

Both clean. Now let me run the full offline suite per project convention to confirm no regressions.

43. user (2026-06-21T00:45:45.198Z)

```
=====
Full test suite: C:\Users\avidu\AppData\Local\Programs\Python\Python313\python.exe -m pytest
-q
obsreport (optional reporting dep): available
=====
..... [ 6%]
..... [ 13%]
..... [ 20%]
..... [ 27%]
..... [ 34%]
..... [ 41%]
..... [ 48%]
..... [ 55%]
..... [ 62%]
..... [ 69%]
..... [ 76%]
..... [ 83%]
..... [ 90%]
..... [ 97%]
..... [100%]
1032 passed in 76.34s (0:01:16)
=====
RESULT: PASS - all tests green
=====
```

44. assistant (2026-06-21T00:45:50.881Z)

Full suite green (1032 passed). Let me confirm the coverage floor and that the new code is covered.

45. user (2026-06-21T00:45:57.966Z)

```
.github/workflows/ci.yml:97:      run: python run_all_tests.py --cov=backend --cov=training
--cov-report=term --cov-fail-under=70
run_all_tests.py:10:    python run_all_tests.py --cov=backend --cov-report=term-missing #
coverage
=== coverage on harness ===
training\gen0\harness.py      129      9   93%   160, 256-264, 268
TOTAL                        129      9   93%
9 passed in 3.49s
```

46. assistant (2026-06-21T00:46:03.679Z)

Coverage floor is 70%; the harness module sits at 93% from this test file alone, with all new `\_git\_sha`/`\_valid\_sha` lines covered. Let me review the full diff before committing.

47. user (2026-06-21T00:46:05.041Z)

```
.gitignore          | Bin 924 -> 1117 bytes
deploy/gcp/README.md | 10 ++++++--
tests/test_training_gen0.py | 56 ++++++
training/gen0/harness.py | 50 ++++++
4 files changed, 111 insertions(+), 5 deletions(-)
=== verify only my commits will be on branch ===
(empty above = clean base)
=== full diff ===
diff --git a/.gitignore b/.gitignore
index 5f68e13..33f8741 100644
Binary files a/.gitignore and b/.gitignore differ
diff --git a/deploy/gcp/README.md b/deploy/gcp/README.md
index 34a1827..17eb9d0 100644
--- a/deploy/gcp/README.md
```

```

+++ b/deploy/gcp/README.md
@@ -149,10 +149,16 @@ The `deploy/digitalocean/` scripts are **provider-agnostic** (they detect
the lo
torch); reuse them on GCP:

```bash
-# from your machine ? ship the repo:
-git archive --format=tar.gz -o /tmp/rally.tgz HEAD
+# from your machine ? ship the repo, BAKING the source commit so the trained bundle is
+# provenance-stamped. A git-archive tarball carries no .git, so training/gen0/harness.py would
+# otherwise record git_sha:"unknown" (seen on the 2026-06-20 L4 run). It reads, in order:
+# live `git rev-parse` ? $RALLY_GIT_SHA ? a GIT_SHA file at the repo root ? "unknown".
+git rev-parse HEAD > /tmp/GIT_SHA
+git archive --format=tar.gz --add-file=/tmp/GIT_SHA -o /tmp/rally.tgz HEAD # GIT_SHA ?
archive root (git ?2.38)
 gcloud compute scp /tmp/rally.tgz rally-gpu:/root/ --zone=asia-south1-a
 gcloud compute ssh rally-gpu --zone=asia-south1-a -- 'mkdir -p ~/rally && tar -xzf ~/rally.tgz
-C ~/rally'
+# (Alternative, if not baking the file: `export RALLY_GIT_SHA=$(git rev-parse HEAD)` on the box
+# in the same shell that runs `python -m backend.worker train ?`.)

on the box:
cd ~/rally
diff --git a/tests/test_training_gen0.py b/tests/test_training_gen0.py
index 9514f6a..3baa5cc 100644
--- a/tests/test_training_gen0.py
+++ b/tests/test_training_gen0.py
@@ -6,6 +6,7 @@ dead-time signal, three rallies per 'video'.
"""

import csv
import json
+import types

import numpy as np
import pytest
@@ -99,6 +100,61 @@ def test_weights_round_trip_and_schema_pin(corpus):
 assert probs.shape == (4,)

+def _git_unavailable(*_a, **_k):
+ """Simulate a run off a `git archive` tarball: no `.git`, so `git rev-parse` cannot run."""
+ raise FileNotFoundError("git: not a repository (tarball)")
+
+
+def test_git_sha_prefers_live_git(monkeypatch):
+ # A real checkout is authoritative ? env var must NOT shadow it (local/CI path unchanged).
+ monkeypatch.setattr(harness.subprocess, "run",
+ lambda *a, **k: types.SimpleNamespace(stdout="a" * 40 + "\n"))
+ monkeypatch.setenv(harness.GIT_SHA_ENV, "b" * 40)
+ assert harness._git_sha() == "a" * 40
+
+
+def test_git_sha_falls_back_to_env_off_tarball(monkeypatch):
+ monkeypatch.setattr(harness.subprocess, "run", _git_unavailable)
+ monkeypatch.setenv(harness.GIT_SHA_ENV, "c" * 40) # case-normalized to lowercase
+ assert harness._git_sha() == "c" * 40
+
+
+def test_git_sha_falls_back_to_baked_file(monkeypatch, tmp_path):
+ monkeypatch.setattr(harness.subprocess, "run", _git_unavailable)
+ monkeypatch.delenv(harness.GIT_SHA_ENV, raising=False)
+ (tmp_path / "GIT_SHA").write_text("d" * 40 + "\n", encoding="utf-8")
+ monkeypatch.setattr(harness, "_REPO_ROOT", str(tmp_path))
+ assert harness._git_sha() == "d" * 40
+

```

```

+
+def test_git_sha_unknown_when_nothing_available(monkeypatch, tmp_path):
+ monkeypatch.setattr(harness.subprocess, "run", _git_unavailable)
+ monkeypatch.delenv(harness.GIT_SHA_ENV, raising=False)
+ monkeypatch.setattr(harness, "_REPO_ROOT", str(tmp_path)) # empty: no GIT_SHA file
+ assert harness._git_sha() == "unknown"
+
+
+def test_git_sha_rejects_garbage_value(monkeypatch, tmp_path):
+ # A non-SHA env value (e.g. a literal "unknown") must fall through, never be stamped.
+ monkeypatch.setattr(harness.subprocess, "run", _git_unavailable)
+ monkeypatch.setenv(harness.GIT_SHA_ENV, "unknown")
+ monkeypatch.setattr(harness, "_REPO_ROOT", str(tmp_path))
+ assert harness._git_sha() == "unknown"
+
+
+def test_manifest_carries_real_sha_from_env_off_tarball(monkeypatch, corpus):
+ """The bug fix, end-to-end: a tarball run (no `.git`) stamps the env SHA into the manifest
+ instead of 'unknown' (regression on gs://.../weights/0.1.0-l4/manifest.json)."""
+ mpath, tmp_path = corpus
+ monkeypatch.setattr(harness.subprocess, "run", _git_unavailable)
+ monkeypatch.setenv(harness.GIT_SHA_ENV, "e" * 40)
+
+ harness.run(mpath, str(tmp_path / "artifacts"), "0.0.9", floor_path=None)
+
+ m = json.loads((tmp_path / "artifacts" / "0.0.9" /
+ "manifest.json").read_text(encoding="utf-8"))
+ assert m["artifact"]["git_sha"] == "e" * 40
+
+
+ def test_sign_test_math():
+ # 5/6 wins -> one-sided p = 7/64 ~ 0.109 (NOT significant ? honesty pinned).
+ wins, n, p = harness.sign_test([1, 1, 1, 1, 1, 0], [0, 0, 0, 0, 0, 1])
+diff --git a/training/gen0/harness.py b/training/gen0/harness.py
+index c012ece..447dc57 100644
+--- a/training/gen0/harness.py
++++ b/training/gen0/harness.py
+@@ -26,6 +26,7 @@ import argparse
+ import hashlib
+ import json
+ import os
+import re
+ import subprocess
+ import time
+ from math import comb
+@@ -47,12 +48,55 @@ CALIBRATION_DEFAULTS = {"smooth": 7, "merge_gap": 1.0, "min_dur": 1.0}
+ THRESHOLD_GRID = [round(t, 2) for t in np.arange(0.2, 0.81, 0.05)]
+
+
+## Repo root = two levels up from training/gen0/harness.py. A `GIT_SHA` file baked here at
+## tarball-build time is the off-box provenance fallback (see _git_sha + deploy/gcp/README.md).
+_REPO_ROOT = os.path.abspath(os.path.join(os.path.dirname(__file__), "..", ".."))
+_SHA_RE = re.compile(r"^[0-9a-f]{7,40}$")
+
+
+##: Env var carrying the source commit SHA out-of-band when the run has no `.git` (a `git
+archive`
+##: tarball ? the GCP-L4 path). Set it on the box, or bake a `GIT_SHA` file into the archive
+root.
+_GIT_SHA_ENV = "RALLY_GIT_SHA"
+
+
+def _valid_sha(s: Optional[str]) -> Optional[str]:
+ """Return the normalized commit SHA if `s` looks like one (7-40 hex chars), else None.
+
+ Guards every provenance source so a stray value (`unknown\n`, a build label) can never

```



```

be
+ stamped as if it were a real SHA.""
+ s = (s or "").strip().lower()
+ return s if _SHA_RE.match(s) else None
+
+
+ def _git_sha() -> str:
+ """Resolve the source commit SHA for the manifest, in priority order:
+
+ 1. live ``git rev-parse HEAD`` ? a real checkout (local dev / CI); authoritative,
+ unchanged.
+ 2. the ``RALLY_GIT_SHA`` env var ? out-of-band injection for a run off a ``git archive``
+ tarball, which carries no ``.git`` (the 2026-06-20 GCP-L4 box stamped ``unknown`` here).
+ 3. a ``GIT_SHA`` file baked into the repo root at tarball-build time.
+ 4. ``"unknown"`` ? nothing available (last resort; never fabricated).
+
+ Steps 2-4 make a remote tarball run provenance-stamped WITHOUT changing the git-present
+ path.
+ """
+ try:
+ return subprocess.run(["git", "rev-parse", "HEAD"], capture_output=True,
+ text=True, timeout=10).stdout.strip() or "unknown"
+ except:
+ out = subprocess.run(["git", "rev-parse", "HEAD"], capture_output=True,
+ text=True, timeout=10).stdout
+ live = _valid_sha(out)
+ if live:
+ return live
+ except Exception:
+ return "unknown"
+ pass
+ env_sha = _valid_sha(os.environ.get(GIT_SHA_ENV))
+ if env_sha:
+ return env_sha
+ try:
+ with open(os.path.join(_REPO_ROOT, "GIT_SHA"), encoding="utf-8") as f:
+ file_sha = _valid_sha(f.read())
+ if file_sha:
+ return file_sha
+ except OSError:
+ pass
+ return "unknown"

def _sha256(path: str) -> str:

```

#### 48. assistant (2026-06-21T00:46:12.707Z)

The diff is clean and the branch base has no sibling commits. The `.gitignore` shows as "binary"  
 ? let me verify my edit landed correctly and the file isn't corrupted.

#### 49. user (2026-06-21T00:46:14.669Z)

```

.gitignore: data
--- encoding probe ---
has NUL: True
first 4 bytes: b'# By'
GIT_SHA present: True

```

#### 50. assistant (2026-06-21T00:46:23.514Z)

The NUL bytes are pre-existing in `.gitignore` (not from my edit ? my entry is present as clean ASCII near the top). Let me confirm the ignore rule actually works and won't let a `GIT_SHA` file slip in.

### 51. user (2026-06-21T00:46:24.913Z)

```
.gitignore:14:/GIT_SHA GIT_SHA
--- rule effective above (exit 0 = ignored) ---
```

### 52. assistant (2026-06-21T00:46:36.289Z)

The ignore rule is effective. Now let me commit with explicit paths (re-verifying HEAD first, per the branching-safety convention).

### 53. user (2026-06-21T00:46:37.602Z)

```
fix/gen0-git-sha-provenance
e6110ce feat(api): GET /api/health + return download_url from /api/compile (M1/P6) (#268)
M .gitignore
M deploy/gcp/README.md
R docs/DECOUPLED_COMPUTE/01-EXISTING-PLANS-AND-GAP.md ->
docs/archives/past_projects/DECOUPLED_COMPUTE/01-EXISTING-PLANS-AND-GAP.md
R docs/DECOUPLED_COMPUTE/02-COMPUTE-MAP-AND-CONTRACT.md ->
docs/archives/past_projects/DECOUPLED_COMPUTE/02-COMPUTE-MAP-AND-CONTRACT.md
R docs/DECOUPLED_COMPUTE/03-PLATFORM-OPTIONS.md ->
docs/archives/past_projects/DECOUPLED_COMPUTE/03-PLATFORM-OPTIONS.md
R docs/DECOUPLED_COMPUTE/04-BUILD-AND-TEST-PLAN.md ->
docs/archives/past_projects/DECOUPLED_COMPUTE/04-BUILD-AND-TEST-PLAN.md
R docs/DECOUPLED_COMPUTE/05-COST-ATTRIBUTION.md ->
docs/archives/past_projects/DECOUPLED_COMPUTE/05-COST-ATTRIBUTION.md
R docs/DECOUPLED_COMPUTE/06-RISKS-AND-OPEN-QUESTIONS.md ->
docs/archives/past_projects/DECOUPLED_COMPUTE/06-RISKS-AND-OPEN-QUESTIONS.md
R docs/DECOUPLED_COMPUTE/07-COMPUTE-ABSTRACTION.md ->
docs/archives/past_projects/DECOUPLED_COMPUTE/07-COMPUTE-ABSTRACTION.md
R docs/DECOUPLED_COMPUTE/DEPLOY_REPORT_GCP_2026-06-20.md ->
docs/archives/past_projects/DECOUPLED_COMPUTE/DEPLOY_REPORT_GCP_2026-06-20.md
R docs/DECOUPLED_COMPUTE/HANDOFF.md -> docs/archives/past_projects/DECOUPLED_COMPUTE/HANDOFF.md
R docs/DECOUPLED_COMPUTE/README.md -> docs/archives/past_projects/DECOUPLED_COMPUTE/README.md
M tests/test_training_gen0.py
M training/gen0/harness.py
?? artifacts/
?? docs/KHELSUTRA_PRODUCT_OVERVIEW.md
```

### 54. assistant (2026-06-21T00:46:57.982Z)

?? Stop ? there are staged `docs/DECOUPLED\_COMPUTE` ? docs/archives/...` renames in the index that are **not mine**. This is the shared-checkout collision the convention warns about: a sibling session staged doc-archive moves into the shared index while I worked. I only `git add`ed my 4 files, but a plain `git commit` would absorb the sibling's staged renames.

Let me verify this and confirm those renames weren't introduced by me.

### 55. user (2026-06-21T00:46:58.628Z)

```
=== staged (cached) name-status ? full index ===
M .gitignore
M deploy/gcp/README.md
R100 docs/DECOUPLED_COMPUTE/01-EXISTING-PLANS-AND-GAP.md
docs/archives/past_projects/DECOUPLED_COMPUTE/01-EXISTING-PLANS-AND-GAP.md
R100 docs/DECOUPLED_COMPUTE/02-COMPUTE-MAP-AND-CONTRACT.md
docs/archives/past_projects/DECOUPLED_COMPUTE/02-COMPUTE-MAP-AND-CONTRACT.md
R100 docs/DECOUPLED_COMPUTE/03-PLATFORM-OPTIONS.md
docs/archives/past_projects/DECOUPLED_COMPUTE/03-PLATFORM-OPTIONS.md
R100 docs/DECOUPLED_COMPUTE/04-BUILD-AND-TEST-PLAN.md
docs/archives/past_projects/DECOUPLED_COMPUTE/04-BUILD-AND-TEST-PLAN.md
R100 docs/DECOUPLED_COMPUTE/05-COST-ATTRIBUTION.md
docs/archives/past_projects/DECOUPLED_COMPUTE/05-COST-ATTRIBUTION.md
R100 docs/DECOUPLED_COMPUTE/06-RISKS-AND-OPEN-QUESTIONS.md
```

```
docs/archives/past_projects/DECOUPLED_COMPUTE/06-RISKS-AND-OPEN-QUESTIONS.md
R100 docs/DECOUPLED_COMPUTE/07-COMPUTE-ABSTRACTION.md
docs/archives/past_projects/DECOUPLED_COMPUTE/07-COMPUTE-ABSTRACTION.md
R100 docs/DECOUPLED_COMPUTE/DEPLOY_REPORT_GCP_2026-06-20.md
docs/archives/past_projects/DECOUPLED_COMPUTE/DEPLOY_REPORT_GCP_2026-06-20.md
R100 docs/DECOUPLED_COMPUTE/HANDOFF.md
docs/archives/past_projects/DECOUPLED_COMPUTE/HANDOFF.md
R100 docs/DECOUPLED_COMPUTE/README.md
docs/archives/past_projects/DECOUPLED_COMPUTE/README.md
M tests/test_training_gen0.py
M training/gen0/harness.py
```

```
=== are the DECOUPLED_COMPUTE moves my doing? check unstaged worktree ===
R docs/DECOUPLED_COMPUTE/01-EXISTING-PLANS-AND-GAP.md ->
docs/archives/past_projects/DECOUPLED_COMPUTE/01-EXISTING-PLANS-AND-GAP.md
R docs/DECOUPLED_COMPUTE/02-COMPUTE-MAP-AND-CONTRACT.md ->
docs/archives/past_projects/DECOUPLED_COMPUTE/02-COMPUTE-MAP-AND-CONTRACT.md
R docs/DECOUPLED_COMPUTE/03-PLATFORM-OPTIONS.md ->
docs/archives/past_projects/DECOUPLED_COMPUTE/03-PLATFORM-OPTIONS.md
```

#### 56. assistant (2026-06-21T00:47:03.973Z)

Confirmed: those renames are a **\*\*sibling session's WIP\*\*** (archiving DECOUPLED\_COMPUTE docs), pre-staged in the shared index ? not mine. I'll do a partial commit restricted to my 4 explicit paths, leaving the sibling's staged renames untouched.

#### 57. user (2026-06-21T00:47:12.346Z)

```
Exit code 1
error: pathspec '-m' did not match any file(s) known to git
error: pathspec '@'
fix(gen0): provenance-stamp the manifest git_sha off a git-archive tarball

A `git archive` tarball carries no `.git`, so `training/gen0/harness.py::_git_sha()`
fell back to "unknown" on a remote box ? the 2026-06-20 GCP-L4 M2 run wrote
git_sha:"unknown" into gs://khelsutra-rally-corpus/weights/0.1.0-l4/manifest.json,
leaving the shipped Gen-0 bundle with no real provenance.
```

Let the SHA be supplied out-of-band. `\_git\_sha()` now resolves in priority order:

1. live `git rev-parse HEAD` (local dev / CI ? authoritative, UNCHANGED path)
2. \$RALLY\_GIT\_SHA env var (out-of-band injection for a tarball run)
3. a `GIT\_SHA` file baked into the repo root at tarball-build time
4. "unknown" (last resort; never fabricated)

Every source is validated against a 7-40 hex-char regex, so a stray value (e.g. a literal "unknown") can never be stamped as a real SHA. Default-OFF-safe: with no env/file and git present, behaviour is byte-identical to before.

- deploy/gcp/README.md §4: bake GIT\_SHA into the archive (`git archive --add-file`)  
or `export RALLY\_GIT\_SHA` on the box, so the trained bundle is provenance-stamped.
- .gitignore: /GIT\_SHA (per-deploy stamp, never a tracked source file).
- tests: 6 new cases ? priority, env/file fallbacks, garbage-rejection, and the end-to-end manifest carrying a real SHA off a (simulated) tarball.

Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>  
@' did not match any file(s) known to git

#### 58. assistant (2026-06-21T00:47:23.053Z)

The `-m` must come before `--`, and that was PowerShell here-string syntax in the Bash tool. Let me write the message to a temp file and commit with `-F`.

#### 59. assistant (2026-06-21T00:47:29.273Z)

fix(gen0): provenance-stamp the manifest git\_sha off a git-archive tarball

A `git archive` tarball carries no `.git`, so training/gen0/harness.py::\_git\_sha() fell back to "unknown" on a remote box ? the 2026-06-20 GCP-L4 M2 run wrote git\_sha:"unknown" into gs://khelsutra-rally-corpus/weights/0.1.0-l4/manifest.json, leaving the shipped Gen-0 bundle with no real provenance.

Let the SHA be supplied out-of-band. \_git\_sha() now resolves in priority order:

1. live `git rev-parse HEAD` (local dev / CI ? authoritative, UNCHANGED path)
2. \$RALLY\_GIT\_SHA env var (out-of-band injection for a tarball run)
3. a GIT\_SHA file baked into the repo root at tarball-build time
4. "unknown" (last resort; never fabricated)

Every source is validated against a 7-40 hex-char regex, so a stray value (e.g. a literal "unknown") can never be stamped as a real SHA. Default-OFF-safe: with no env/file and git present, behaviour is byte-identical to before.

- deploy/gcp/README.md §4: bake GIT\_SHA into the archive (git archive --add-file) or `export RALLY\_GIT\_SHA` on the box, so the trained bundle is provenance-stamped.
- .gitignore: /GIT\_SHA (per-deploy stamp, never a tracked source file).
- tests: 6 new cases ? priority, env/file fallbacks, garbage-rejection, and the end-to-end manifest carrying a real SHA off a (simulated) tarball.

Co-Authored-By: Claude Opus 4.8 <noreply@anthropic.com>

#### 60. user (2026-06-21T00:47:29.290Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\.git\COMMIT\_MSG\_gitsha.txt (file state is current in your context ? no need to Read it back)

#### 61. user (2026-06-21T00:47:35.909Z)

[fix/gen0-git-sha-provenance fc862a9] fix(gen0): provenance-stamp the manifest git\_sha off a git-archive tarball

4 files changed, 111 insertions(+), 5 deletions(-)  
=== verify branch vs origin/master holds ONLY my commit ===  
fc862a9 fix(gen0): provenance-stamp the manifest git\_sha off a git-archive tarball  
=== sibling renames still staged & untouched? ===  
M docs/README.md  
R100 docs/DECOUPLED\_COMPUTE/01-EXISTING-PLANS-AND-GAP.md  
docs/archives/past\_projects/DECOUPLED\_COMPUTE/01-EXISTING-PLANS-AND-GAP.md  
R100 docs/DECOUPLED\_COMPUTE/02-COMPUTE-MAP-AND-CONTRACT.md  
docs/archives/past\_projects/DECOUPLED\_COMPUTE/02-COMPUTE-MAP-AND-CONTRACT.md

#### 62. assistant (2026-06-21T00:47:44.111Z)

Clean: the commit changed exactly my 4 files, the branch holds only my commit over `origin/master`, and the sibling's staged renames remain untouched in the index for them. Now let me push and open the PR.

#### 63. user (2026-06-21T00:47:46.557Z)

remote:  
remote: Create a pull request for 'fix/gen0-git-sha-provenance' on GitHub by visiting:  
remote: <https://github.com/Khelsutra/badminton-highlight-indexer/pull/new/fix/gen0-git-sha-provenance>  
remote:  
branch 'fix/gen0-git-sha-provenance' set up to track 'origin/fix/gen0-git-sha-provenance'.  
To <https://github.com/Khelsutra/badminton-highlight-indexer.git>  
\* [new branch] fix/gen0-git-sha-provenance -> fix/gen0-git-sha-provenance

#### 64. assistant (2026-06-21T00:48:01.037Z)

## Problem

The Gen-0 weights bundle written by ``worker train`` recorded ``git_sha: "unknown"`` in ``weights/<version>/manifest.json`` whenever the run happened on a remote box ported via ``git archive`` ? the tarball has no ``.git``, so ``training/gen0/harness.py::_git_sha()``'s ``git rev-parse HEAD`` failed and fell back to ``.unknown```.

Observed on the **2026-06-20 GCP-L4 M2 run** ?  
``gs://khelsutra-rally-corpus/weights/0.1.0-l4/manifest.json`` shipped with **no real provenance** (the one immutable field that ties a served model back to its source commit).

## Fix

Let the SHA be supplied **out-of-band** so the bundle is provenance-stamped even off a tarball. ``.git_sha()`` now resolves in priority order:

1. **live ``git rev-parse HEAD``** ? a real checkout (local dev / CI); authoritative, **path unchanged**
2. **``${RALLY_GIT_SHA}`` env var** ? out-of-band injection for a ``git archive`` tarball run
3. a **``${GIT_SHA}`` file** baked into the repo root at tarball-build time
4. **``.unknown```** ? last resort; never fabricated

Every candidate is validated against a ``.^[0-9a-f]{7,40}$`` regex, so a stray value (e.g. a literal ``.unknown```, a build label) can never be stamped as if it were a real SHA.

**Default-OFF-safe:** with no env var / no ``.GIT_SHA`` file and ``git`` present, behaviour is byte-identical to before (the local path is untouched).

## Changes

- ``training/gen0/harness.py`` ? resolution chain + ``.valid_sha()`` guard + ``.GIT_SHA_ENV`` constant.
- ``deploy/gcp/README.md`` \$4 ? bake the SHA into the tarball (``git archive --add-file=tmp/GIT_SHA``)  
or ``export RALLY_GIT_SHA=$(git rev-parse HEAD)`` on the box, so the trained bundle is stamped.
- ``.gitignore`` ? ``.GIT_SHA`` (a per-deploy stamp, never a tracked source file).
- ``tests/test_training_gen0.py`` ? **6 new cases**: live-git priority, env fallback, baked-file fallback, nothing-available ? ``.unknown```, garbage-rejection, and the **end-to-end manifest** carrying a real SHA off a (simulated) tarball.

> Note: this is forward-looking ? the already-shipped ``.0.1.0-l4`` manifest still reads  
> ``.unknown```; re-stamping it would require a re-run (owner-gated, not in this PR).

## Verification

- ``python run_all_tests.py`` ? **1032 passed** (1026 + 6 new).
- ``ruff check`` clean; ``mypy training/gen0/harness.py`` clean.
- Coverage: ``training/gen0/harness.py`` at **93%** from its test file alone (CI floor 70%); all new lines covered.

? Generated with [Claude Code](https://claude.com/claude-code)

### 65. user (2026-06-21T00:48:01.051Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\.git\PR\_BODY\_gitsha.md (file state is current in your context ? no need to Read it back)

### 66. user (2026-06-21T00:48:11.003Z)

<https://github.com/Khelsutra/badminton-highlight-indexer/pull/269>

### 67. assistant (2026-06-21T00:48:16.087Z)

PR created: [#269](https://github.com/Khelsutra/badminton-highlight-indexer/pull/269). Let me update the live current-state memory per the micro-checkpoint convention.

## 68. user (2026-06-21T00:48:17.128Z)

```
1 ---
2 name: current-state
3 description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
4 only on request). What just landed, what's next, open decisions."
5 metadata:
6 node_type: memory
7 type: project
8 originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
9 ---
10 # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12 > **Scope of this file = LIVE only:** the current handoff + *today's* checkpoints + open
13 PRs/decisions. Keep it read-first-short (?1 page). Older checkpoints are archived in [[current-
14 state-archive]] ? when today's checkpoints age out (next clean-slate day), move them there,
15 don't let them pile up here.
16
17 **? HANDOFF (2026-06-20) ? KHELSUTRA.GURU brand site: shipped, live, portable. Resume
18 the product threads below.** Separate track from the indexer rally-detection handoff that
19 follows. Full durable record + ramp-up: [[khelsutra-domain-launch]]. Repo =
20 **`avidullu/khelsutra`** (NOT this indexer repo).
21
22 - **? START HERE (2026-06-20, latest ? big session, ALL MERGED, NO open PRs).** Per-
23 rally **MVP/M0 COMPLETE** + a full product/promo wave. **khelsutra master `a7214ad`** . **engine
24 (Khelsutra/badminton-highlight-indexer) master has M1 [#268]**.
25
26 - **Per-rally MVP (khelsutra `web/` SPA):** P3 `parseQuery()` query bar [#28]
27 (length/order/count/shot_count; greys shot-type/player/score "not available yet" ? never faked;
28 **13 adversarial-review findings folded in**) . P4 `SelectionFooter`+honest `min_shot_count`
29 warning [#30] (`web/src/lib/floor.ts` mirrors `main.py:362-383`; no-shot_count exempt; all-below
30 ? Build DISABLED) . P5 real load `?video=<id>` + Build?`compileReel`?`<video>` preview+download
31 [#31]. Full obs (`query.*`/`build.*`/one cid). **Then: lead-with-LENGTH, shot-count flagged
32 EXPERIMENTAL** [#33] (cards show `m:ss long`; amber "experimental" chip; duration-led examples;
33 badge `P5`?`alpha`; parseQuery still accepts shot-count, de-featured). Suite **75 green**.
34
35 - **Engine M1 [#268] (GCP/org repo, built in an ISOLATED git worktree off
36 origin/master so the concurrent session's checkout was untouched):** `GET /api/health`
37 ({status,sport,default_segmenter}) + `POST /api/compile` now returns **`download_url`**
38 (additive; `filepath` unchanged). Full engine suite **1019 passed/7 skipped**, ruff+mypy clean,
39 CI green (coverage floor incl. torch). *(SPA can adopt `download_url`/offline-banner as a
40 follow-up.)*
41
42 - **GCS round-trip (reuse artifacts, CPU/no-GPU):** pulled
43 `mahadevpura_smoke_180s.mp4` + intervals from the bucket ? **stitched a curated top-10-longest
44 highlight** (ffmpeg) ? **put it back** at `gs://khelsutra-rally-
45 corpus/highlights/curated_top10_longest.mp4` (61s).
46
47 - **3 screencasts (Playwright?webm?ffmpeg):** `promo_cuj.mp4` (old UI, real footage) .
48 **`full_cuj_real.mp4`** (NEW duration-first UI + REAL curated reel ? the definitive full-CUJ) ?
49 both in `gs://khelsutra-rally-corpus/promo/` (real footage ? bucket, NOT public repo) .
50 **`site/public/video/tool-cuj.mp4`** (NEW UI, **footage-free** branded-placeholder reel ? safe
51 to self-host). Local copies in `badminton-highlight-indexer/artifacts/promo/` (gitignored).
52
53 - **Site (excite-to-join) [#34/#35]:** after sign-up the reveal now shows **(1) the
54 tool in action** (self-hosted footage-free `tool-cuj.mp4`) **+ (2) a ~30s reel preview**
55 (`REEL_SECS` 90?30, gated nocookie YouTube ? real faces stay in the unlisted reel, not the
56 repo); `revealAfterSignup`; coherent CTA/teaser copy; `server.js` serves `.mp4` as `video/mp4`;
57 `*.mp4`/`*.gif` marked binary. Site suite **34 green** ; **droplet re-synced + parity GREEN**
58 (index+mp4 byte-identical CF?VM:8787).
59
60 - **? GPU note:** could NOT spin a fresh GPU VM ? `GPUS_ALL_REGIONS=1` slot held by
61 the concurrent session's `rally-gpu` M2 training (did NOT disturb). Validated CUJ against the
62 first-L4-run REAL artifacts instead. A fresh live `/api/process` GPU inference stays
63 **deferred** until the GPU frees (low-risk existing path). **Owner observation ? [[machine-
64 pooling-idea]]** (pool GPUs vs spin/kill per task).
65
66 - **Owner decision flags (PUBLIC promo):** `full_cuj_real.mp4`/`promo_cuj.mp4` show
67 real court footage (faces) ? for public use beyond the gated site, confirm. The footage-free
68 `tool-cuj.mp4` is safe.
69
70 - **OPEN owner-gated (unchanged):** appliance $10 decisions; move khelsutra site repo
71 ? Khelsutra org (needs Cloudflare GitHub-App reconnect). **Next options:** SPA adopt
```

`download\_url`+offline-banner (uses M1); engine P7 (`/api/videos/{id}/rallies` alias); produce a polished public promo cut.

24 - **SESSION WRAP** (2026-06-20 late) ? site fully LIVE on Cloudflare + the D0 droplet twin (`/srv/khelsutra-site`), byte-in-sync (parity green); PRs #1?#26 merged, NO open PRs, master `9f0942b`. **This session shipped: \*\*per-rally MVP P1+P2\*\*** (`web/` SPA: Vite/React/TS/Tailwind + typed engine client + `RallyGrid` + `useRallySelection`; 16 vitest; **\*\*P3 now done ? see START HERE above\*\***) . **\*\*3 tracked design docs merged\*\*** (`PER\_RALLY\_SELECTION.md` IN-PROGRESS, `LOCAL\_APPLIANCE\_ARCHITECTURE.md`, `ANNOTATE\_AND\_TRAIN.md`) . live **\*\*`own-model`\*\*** page (real annotation-tool screenshots [no people] in `site/public/img/`, sample reel **\*\*gated\*\*** behind request-access, "default works out of the box / BYO-training **\*\*optional\*\***" framing, **\*\*? KhelSutra home\*\*** back-link) . **\*\*provider-portable\*\*** `site/scripts/provision-vm.sh`+`teardown-vm.sh` (LIVE-tested: fresh D0 VM ? operational ? state-export ? destroy) + the **\*\*git -c core.autocrlf=false archive\*\*** CRLF fix . **\*\*homepage sample reel\*\*** gated post-signup (random-start `youtube-nocookie` + 90s auto-close) . **\*\*`rally-annotator` TRANSFERRED ? `Khelsutra` org\*\*** (links repointed). **\*\*Site CI = 32 green\*\*** (`npm test`) + scheduled live-parity workflow. **\*\*GPU deferred to the owner's Windows+WSL box\*\*** (D0 has no GPU SKUs + GCP quota=0; no Gemini e2e per owner). **\*\*OPEN owner threads:\*\*** per-rally **\*\*P3+\*\***; appliance **\*\*\$10 decisions\*\*** (edge-auth / retention / per-video cost cap / GPU host / P5 bucket-worker bridge); **\*\*move sister repos ? `Khelsutra` org as PRIVATE\*\*** (verified feasible ? free org allows 10000 private repos + collaborators, owner stays admin; ENGINE-repo move needs coordinating w/ active sessions ? see `khelsutra/NEXT\_STEPS.md` "Org & repos"); optional VLC-CSV annotate walkthrough screenshots. ? Droplet STILL a shared testbed (only `/srv/khelsutra-site` touched; never the engine `~/rally/MinIO/GCS`). **\*\*Ramp-up:\*\*** this block ? [[khelsutra-domain-launch]] ? `khelsutra/NEXT\_STEPS.md` ? the 4 design docs (`PER\_RALLY\_SELECTION` / `LOCAL\_APPLIANCE\_ARCHITECTURE` / `ANNOTATE\_AND\_TRAIN` / `PORTABILITY`). Droplet SSH: `~/ssh/khelsutra\_droplet` ? `root@168.144.126.176`; sync recipe in `PORTABILITY.md`.

25 - **ORG CONSOLIDATION DONE** (2026-06-20 late): **\*\*transferred \*\***badminton-highlight-indexer` (engine), `sports-data-collector`, `sports-obsreport` ? the `Khelsutra` GitHub org **\*\*** (all stay **\*\*private\*\***; verified `Khelsutra`<repo>` private). Safe: no open PRs, no forks, already-private (no visibility change); GitHub **\*\*redirects git ops\*\*** so active quality-iteration sessions' push/pull/CI keep working (local files untouched). **\*\*Local remotes repointed\*\*** to `https://github.com/Khelsutra/<repo>.git` (engine ls-remote verified). **\*\*`rally-annotator` stays PUBLIC\*\*** (OSS MIT plugin the site links to ? do NOT make private). **\*\*`khelsutra` (site repo) NOT moved\*\*** ? a transfer breaks the **\*\*Cloudflare Workers-Builds git connection\*\*** (auto-deploy) until the CF GitHub App is installed on the `Khelsutra` org + the Worker reconnected (owner dashboard step) ? **\*\*pending owner\*\***. Engine harness/memory path UNCHANGED (keyed to the local dir name `badminton-highlight-indexer`, not the GitHub URL). ? Could not read org Actions policy (gh token lacks `admin:org`); engine CI history intact at the new path (last run = success) + free orgs default Actions-on ? verify CI on the next engine PR; enable in org settings if needed.

26 - **DONE + LIVE** this session (PRs #1?#8, all merged, repo clean): **\*\*owner bought khelsutra.guru ? built & launched end-to-end. On \*\*Cloudflare\*\*** (Worker `hidden-king-khelsutra` + D1 `khelsutra-access`; **\*\*git-connected ? merge `master` auto-deploys\*\***; CI runs `npm test` = 20 green): landing page + `/capture` checklist + OG image + a **\*\*segmented Request-Access form\*\*** (academy/individual + GPU + storage Qs + free text) ? persists to D1 **\*\*and emails a summary to avi.dullu@gmail.com per registration\*\*** (Email Routing send binding ? VERIFIED live in inbox). Valid SSL; `hello@` Email Routing verified.

27 - **PORTABLE + proven\*\***: identical form runs on a dependency-free **\*\*Node + SQLite\*\*** host (`site/server.js`), **\*\*live + browsable on the owner's D0 droplet ? http://168.144.126.176:8787\*\*** (systemd `khelsutra-site`, isolated `/srv/khelsutra-site`; SSH key `~/ssh/khelsutra\_droplet`). Both-hosts compat ENFORCED in CI (`site/src/server.test.js` boots the real server). ? Droplet is a SHARED storage testbed ? another session runs MinIO 9000/9001 + GCS emu 9023 + `/root/do\_gcs\_`; STAY CLEAR. Docs: `docs/REQUEST\_ACCESS\_FORM.md`, `docs/PORTABILITY.md`, `NEXT\_STEPS.md`.

28 - **?? PENDING / next threads** (owner ideas this session): **\*\* (1) \*\*Marketing video\*\*** ? use Grok/Gemini image-gen ? a short video themed for **\*\*serious enthusiasts\*\*** showing the whole record?reel process ? embed on the site. **\*\* (2) \*\*Interactive per-rally selection\*\*** (big product feature) ? the engine detects rallies ? present them in a UI so the user **\*\*picks which rallies go into the highlight\*\*** (dynamic, NOT a fixed setting). ? **\*\*Design doc MERGED ? [khelsutra#11](https://github.com/avidullu/khelsutra/pull/11); owner APPROVED ? decisions D8?D11 locked (status IN PROGRESS). ? CI regression-guards MERGED ? [khelsutra#12](https://github.com/avidullu/khelsutra/pull/12): `site/src/deploy-config.test.js` (pins wrangler.toml deploy invariants ? the #5 name-drift class), `site/src/assets.test.js` (static-site content contract), `.github/workflows/parity.yml` (scheduled+dispatch live-parity, NON-blocking); `npm test`?`node --test` discovery; suite 20?28 green; live-parity dispatched ? BOTH hosts byte-in-sync. ? **\*\*P1 MERGED ?****

[khelsutra#14](https://github.com/avidullu/khelsutra/pull/14)\*\* (squash `aac7e0c`): `web/` SPA scaffold (Vite6+React19+TS+Tailwind4) + \*\*typed engine API client\*\*

(`listRallies`/`compileReel`/`highlightUrl`/`getConfig`+`minShotCount`, grounded+verified; honest types flag `server/receiver/events` as motion fabrications) + \*\*correlation-id structured logging\*\* (`web/src/lib/log.ts`, observability seed, loud failures) + \*\*8 vitest green\*\* + new \*\*web` CI job\*\* (npm ci?typecheck?test?build, gated lockfile); live-verified renders in-browser with ZERO console errors. Isolated (no README/NEXT\_STEPS edits ? no #13 conflict). ? \*\*P2 DONE ?

[khelsutra#21](https://github.com/avidullu/khelsutra/pull/21):\*\* `RallyGrid` (honest fields only) + `useRallySelection` hook over pure `lib/selection.ts`+`lib/format.ts` (default ALL-on per D8); 16 vitest green; live-verified (deselect ? footer recompute 6/1:22?5/0:53). \*\*NEXT = P3\*\* `parseQuery()` ? P4 footer+`min\_shot\_count` warning ? P5 build?compile?download. All on existing endpoints.\*\* Built via a multi-agent design Workflow (research real engine contract ? 3 approaches ? synthesize) with \*\*every load-bearing claim personally re-verified vs engine source\*\*. Key finding: \*\*MVP needs ZERO new engine endpoints\*\* ? rides existing `POST /api/query` ? select `segment\_ids` ? `POST /api/compile` ? `GET /api/highlights`. Recommended hybrid: selectable rally \*\*grid\*\* (workhorse) + \*\*deterministic `parseQuery()`\*\* (no LLM; length/count/order/shot\_count ONLY) writing into ONE shared selection set + sticky compile footer. Honest scope: shot/player/score = roadmap (no column/detector);

`server`/`receiver`/`events` = `motion`-only fabrications ? hidden; default `gemini` writes real-ish `shot\_count`; per-rally \*preview\* blocked until a source/clip stream endpoint lands (deferred milestone = biggest UX-vs-effort lift). Owner decisions RESOLVED (D8?D11, 2026-06-20): (a) default selection \*\*ON\*\* (curate-by-removal); (b) preview-before-select \*\*NOT\*\* a ship-blocker (text-metadata MVP; preview = P9); (c) \*\*job-tethered\*\* MVP (no `GET /api/videos`); (d) `min\_shot\_count` = \*\*warn, don't override\*\* (override deferred ? Launch Report). Doc = `khelsutra/docs/PER\_RALLY\_SELECTION.md`. Pairs with the "ask for a reel" conversational builder already teased on the site. (3) productionize droplet (TLS + `vm.khelsutra.guru`), VM-path SMTP email (currently noop), Always-Use-HTTPS/HSTS, Web Analytics token, `/capture.html`?`/capture` footer link, delete any orphan `khelsutra-site` Worker, clean CF D1 test rows (`deploy-test@`/`email-test@`). Revoke the droplet SSH key when done. [[paper-coauthor-aim]]

[monetization-plan]

29 - \*\*? APPLIANCE DESIGN ? doc SHIPPED for review ?

[khelsutra#13](https://github.com/avidullu/khelsutra/pull/13) (DRAFT, owner-gated; CI green) ? "locally-installed, UI-powered rally indexer + reel generator" (KhelSutra Studio).\*\* Owner target (HEADLINE CUJ): a \*\*hosted website\*\* + a \*\*NEW GPU-enabled machine (provider-agnostic: DO GPU droplet / GCE GPU VM / Cloud Run+GPU)\*\* running the decoupled e2e; from the UI the user picks an input ? \*\*Google Drive / bucket URI (S3/GCS/DO Spaces) / local upload?bucket\*\* ? then \*\*generates + watches + downloads\*\* highlights \*\*completely from the UI\*\*. Framed as a \*\*deployment PROFILE of the decoupled-compute arch\*\* (engine `backend/storage/` StorageBackend + `python -m backend.worker {infer|train}` bucket pipeline; PRs #246?#254 on engine master, 936 green; \*\*the CPU droplet 168.144.126.176 already hosts BOTH the site `~/srv/khelsutra-site` AND the engine `~/rally`, GPU MOCKED via stub). ? \*\*OWNER PRIORITY (2026-06-20): MOVE TO A GPU-ENABLED MACHINE ASAP ? provider-agnostic: DO GPU droplet / GCE GPU VM / Cloud Run+GPU\*\* (storage stays the S3/GCS bucket; GCS = storage, GCE/Cloud Run = compute ? do NOT conflate). Critical-path deps that are NOT mine to do unilaterally: (owner) stand up the GPU compute target + provide the cloud API token/creds ? DO runbook = engine `docs/SETUP\_NEW\_MACHINE.md` \$2/\$9 + `deploy/digitalocean/`; \*\*Cloud Run / GCE need the `backend.worker` CONTAINERIZED (likely a GAP; WASB's torch-1.11 conda env makes the image non-trivial) ? verify\*\*. Cloud Run+GPU (scale-to-zero) is the strong fit for the per-video-billing economics (engine DECOUPLED\_COMPUTE #246). (engine session) build \*\*`NativeWasbRunner` (M3.5)\*\* for real (non-mock) WASB. The \*\*UI/appliance work runs in PARALLEL\*\* (rides the existing FastAPI: chunked upload, `/api/process`?`/api/jobs`, `/api/query`, `/api/compile`, `/api/highlights`). ? Engine API has \*\*NO auth (CORS ``)\*\* ? \*\*EDGE AUTH required before any public exposure\*\*. ? Google \*\*Drive ingest may be a `gdrive` STUB\*\* in `backend/storage/` (#249) ? verify before promising. Doc ? `khelsutra/docs/LOCAL\_APPLIANCE\_ARCHITECTURE.md` (PR pending). The merged per-rally selection (#11) = ONE screen of this operator console. CUJ/live-test plan is a first-class doc section (owner asked re: live CUJ replay). \*\*RECOMMENDATION (B+C hybrid):\*\* reverse-proxied \*\*UNIFIED ORIGIN\*\* ? `web/` SPA behind edge auth, `/api/` ? engine `127.0.0.1:8000` (CORS ``` moot; gate = tenancy boundary; bind verified `main.py:638`) ? driving the \*\*EXISTING\*\* FastAPI loop (upload?process?jobs?query?compile?highlights) with `storage=local`; \*\*compute-as-a-setting\*\*, bucket/remote-GPU worker = reserved scale path. \*\*Verified-honesty refinements:\*\* native WASB PARTIALLY wired (`fusion\_hybrid` builds `NativeWasbRunner` :86; `trajectory\_hybrid` refuses :57-63) + owner-byte-parity-gated; `ai\_max\_workers` default=\*\*5\*\* not 1; chunked-upload endpoints EXIST (`uploads.py:64/89/123`) but `app.js` never calls them (client-wiring only). \*\*NET-NEW engine endpoints to COORDINATE\*\* with the decoupled sessions: `/api/capabilities`, `/api/videos`, `/api/reels`, `/api/health`. \*\*\$10 owner decisions PENDING:\*\* edge-auth (Caddy



basicauth vs cloudflared+Access), retention defaults, per-video cost ceiling, GPU host choice, P5 bridge ownership. **\*\*Observability = first-class \$5\*\*** (correlation-id e2e, structured logs per path, loud failures, health/metrics) per `[[feedback-launch-quality-bar]]`. Doc = ``khelsutra/docs/LOCAL_APPLIANCE_ARCHITECTURE.md``. **\*\*DELIVERY LOCKED (D10, 2026-06-20):\*\*** self-hosted **\*\*webserver + UI the user spins up\*\*** (NOT a desktop app; supersedes ``DESKTOP_APP_PLAN.md``); user points at local storage+GPU/CPU OR cloud, UI drives the tool. **\*\*PROVISIONING ACCESS (2026-06-20):\*\*** ``doctl`` **\*\*INSTALLED v1.162.0\*\*** (winget ``DigitalOcean.Docker``; at ``%LOCALAPPDATA%\Microsoft\WinGet\Links\doctl.exe``, on PATH) ? **\*\*doctl AUTHED\*\*** (team "Khelsutra Guru", droplet-limit 3; 1 used = ``claude-do-rally-test`` blr1 CPU @168.144.126.176). ? **\*\*GPU BLOCKED ON BOTH PROVIDERS (verified 2026-06-20):\*\*** DO catalog has **\*\*NO GPU sizes\*\*** (17 CPU-only sizes; DO GPU is access-gated + region-limited to NYC/TOR/ATL, not blr1) AND **\*\*GCP `GPUS\_ALL\_REGIONS` quota = 0\*\*** (Compute API enabled). Both need an **\*\*owner-initiated, provider-approved access/quota request\*\*** ? NOT CLI-doable (GCP quota-increase = fastest realistic path since gcloud is authed as avi.dullu@gmail.com, project ``gen-lang-client-0306026826``). ? **\*\*DO NOT DELETE the old droplet\*\*** (``claude-do-rally-test``) ? it's the SHARED decoupled-session testbed (``~/rally`` engine, MinIO 9000/9001, GCS emu 9023, validated DO Spaces e2e); deleting destroys their env. Site-migration + droplet-deletion is MOOT until a GPU VM exists anyway. ? **\*\*Real product e2e (upload?index?reel) is testable NOW via Gemini (no GPU needed)\*\*** ? GPU is only for the owned/native-WASB model + the train-your-own-model USP (also pending the engine session's ``NativeWasbRunner`` M3.5). ? **\*\*Annotate+train USP design SHIPPED ? [khelsutra#15](https://github.com/avidullu/khelsutra/pull/15)\*\*** (DRAFT, owner-gated; CI green). Plan: ONE marketing page ``site/public/own-model.html`` (Annotate/Train/Setup sections, clones ``capture.html``, links MIT rally-annotator) + a design-tracked owner-gated SPA correction screen (NO "Train now" button ? training is offline ``backend/worker train`` CLI). VERIFIED: annotator CSV (``rally_number,start_time,end_time,ending_reason,sport,shots_count``) ingests with **\*\*NO transform\*\*** (synth ``DictReader row["start_time"]``, ``stub_runner.py:139`` + ``gt_loader``); ONE prereq = **\*\*MD5+stem video?label join\*\*** (never re-encode after annotating). Honest: annotator real+shipping (v1.6.4 MIT VLC plugin); train half = open CLI + roadmap (GPU-free=CPU **\*\*mock\*\*** only; real needs GPU + in-flight native WASB; ship-gate ``gate_verdict`` ``harness.py:98`` refuses commercial ship; ``label_source=human`` + ``no_paid_llm_training_labels`` attested :178/:189-192); annotator repo has **\*\*ZERO images ? all screenshots TO-CAPTURE\*\*** (consented, no minors, NEVER fabricate); USP = ownership/licensing/privacy/consent/minors-excluded, **\*\*NOT accuracy\*\*** (no benchmark vs paid LLM). **\*\*GPU DECISION (owner 2026-06-20):\*\*** use the **\*\*Windows+WSL GPU box\*\*** for real testing; **\*\*DEFER the cloud GPU VM\*\***; **\*\*NO Gemini e2e now\*\***; **\*\*NO droplet migration/deletion\*\***. Real WASB e2e ? owner's box when run deliberately. **\*\*#13 (appliance) + #15 (annotate/train) MERGED\*\*** (owner-approved). ? **\*\*PROVIDER-PORTABILITY now first-class + LIVE-TESTED ? [khelsutra#16](https://github.com/avidullu/khelsutra/pull/16) MERGED:\*\*** ``site/scripts/provision-vm.sh`` (fresh Ubuntu VM on ANY provider ? ``git archive`` push [private-repo-safe, no secret on VM] ? Node24 + systemd + firewall + verify) + ``teardown-vm.sh`` (EXPORT SQLite state to local backup BEFORE destroy ? clean resume via ``RESTORE_DB``) + PORTABILITY runbook + CI ``bash -n`` parse-check. **\*\*PROVEN on a throwaway DO droplet (then destroyed):\*\*** bare ? operational ? public HTTP 200 ? state-export ? re-provision-with-RESTORE\_DB preserved the row (``rows:2``) ? droplet+temp-key removed. So destroy/migrate-across-providers is now safe + reproducible (owner principle: "deleting is fine IF we can resume cleanly via scripts"). ? Still **\*\*DO NOT delete the shared `claude-do-rally-test` droplet\*\*** without coordinating the decoupled session (their ``~/rally`` engine env reproduces via the ENGINE repo's ``SETUP_NEW_MACHINE.md``, not mine). ? **\*\*UI PROPOSAL rendered\*\*** (visualize mockup): an **\*\*"Own your model" section for the EXISTING landing page ? Annotate-in-VLC (live) / Train-your-own (in-build, honest) / Yours+private ? links the MIT annotator + a `own-model` setup guide; ? `own-model` page LIVE ? [khelsutra#22](https://github.com/avidullu/khelsutra/pull/22)\*\*** (``site/public/own-model.html``: Annotate=live / Train=in-build / Setup; honest framing, GitHub-linked, **\*\*placeholder\*\*** screenshots; ``own-model`` alias + nav/footer cross-links; deployed to BOTH hosts, parity green; real screenshots = owner-gated **\*\*P5\*\***). ? **\*\*own-model reel REUSED as click-to-play ? [khelsutra#24](https://github.com/avidullu/khelsutra/pull/24):\*\*** owner asked to reuse the existing reel rather than capture screenshots ? the video is a REEL (engine output), NOT a shot of the annotator UI, so it can't replace a tool screenshot, but it's embedded as a **\*\*branded click-to-play\*\*** (ink poster + play button, NO footage/faces until tap ? ``youtube-nocookie`` IFrame-API player, **\*\*random start ?90s before end\*\***). Footage hidden until tap (privacy-consistent w/ homepage); verified live both hosts (``facade=4``, static-iframe=0, thumbnail=0, parity green md5 ``90de6a9e``). ? **\*\*SUPERSEDED by [khelsutra#25](https://github.com/avidullu/khelsutra/pull/25) ? more coherent layout (owner call):\*\*** the reel (shows players) is now GATED in ONE place only (homepage request-access); the own-model page no longer embeds it. Instead own-model shows the **\*\*real annotation-T00L screenshots\*\*** (owner provided 2, NO people: ``site/public/img/annotate-enable.png`` = VLC View?Rally Annotator, ``annotate-tool.png`` = the Mark START/END/Save dialog) + a one-line

\*\*\*request access to see a sample output\*\*\* pointer + a \*\*\*you don't have to train anything ? a ready-made default works out of the box; BYO-training is OPTIONAL\*\*\* note (keeps non-tech users engaged). Tests assert screenshots-present + no-reel-embed + default/optional framing. Verified live BOTH hosts: images HTTP 200 image/png, md5 `b6350339` identical local/droplet/CF (binary integrity through git-archive), own-model `iframe=0/youtube=0`, parity green. (Reel still lives gated on the homepage: random-start nocookie + 90s auto-close after signup.) ? \*\*rally-annotator ? Khelsutra org:\*\* owner created the `Khelsutra` GitHub org; transfer is FEASIBLE (owner = org admin, no name conflict) ? ? \*\*DONE: TRANSFERRED\*\* to `Khelsutra/rally-annotator` (org owns it; old `avidullu/?` URL auto-redirects). Site/doc links repointed ?

[khelsutra#23](https://github.com/avidullu/khelsutra/pull/23); deployed to BOTH hosts, `own-model` parity green (md5 `c0af70fb`). Repo `git remote` for any local rally-annotator checkout should be updated to the Khelsutra URL. GPU still deferred to the Windows+WSL box (no cloud GPU; no Gemini e2e). ? \*\*DEMO REEL ? SIGN-UP-GATED + on BOTH hosts (#17 open ? #18 gated ? #19 fix).\*\* The owner's UNLISTED reel `ajTCZ7pJw\_o` is NOT shown openly; instead a "Sign up to see it in action" teaser + on a successful request-access submit it reveals a `youtube-nocookie` autoplay embed that \*\*AUTO-CLOSES in 90s\*\* (`#reel-reveal` + live countdown; [khelsutra#18](https://github.com/avidullu/khelsutra/pull/18) merged; verified live submit/reveal on the Node host; `assets.test.js` asserts GATED-not-open; suite 29 green).

\*\*Deployed to BOTH:\*\* Cloudflare auto-deployed; \*\*droplet `/srv/khelsutra-site` twin re-synced to master ? PARITY GREEN\*\* (`OK / 231dd90` . `OK /capture 2ada2cf`; access.db + self-contained Node 26 preserved; only `srv/khelsutra-site` touched). ? \*\*CRLF-drift bug found live + fixed ? [khelsutra#19](https://github.com/avidullu/khelsutra/pull/19):\*\* Windows `core.autocrlf=true` overrides `.gitattributes eol=lf` in `git archive` ? CRLF on the VM ? byte parity drift vs CF's LF; `provision-vm.sh` + `parity-check.sh` now pin `git -c core.autocrlf=false archive` (LF cross-platform; no-op on Linux). Privacy now stronger: the clip only plays after signup (not for every visitor/crawler); YT stays owner-controlled. ? \*\*#20 (random start + join FAQ):\*\* the reel now starts at a \*\*RANDOM point ?90s before the end\*\* (YT IFrame Player API: `getDuration()` ? random `[0,dur-90]` ? `seekTo`; falls back to 0 if dur unknown; verified live = 35s then 20s across two sign-ups) so people don't all see the same chunk; the \*\*\*"How do I join?" FAQ\*\* now points to the request-access form (`#join`, no `mailto`) ? consistent with the actual signup. Both hosts re-synced LF-safe ? \*\*parity GREEN\*\* (`848312?`); site suite \*\*30 green\*\*.

[[feedback-launch-quality-bar]] \*\*`gcloud` IS authed\*\* here (avi.dullu@gmail.com, project `gen-lang-client-0306026826` ? likely needs GPU quota + billing verification before GCE/Cloud Run). Owner prefers \*\*DO first\*\* for GPU. Creating paid GPU infra = confirm-first every time; I propose exact cmd + hourly cost + teardown, owner OKs. [[monetization-plan]] [[khelsutra-domain-launch]]

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33 \*\*? HANDOFF (2026-06-20) ? DECOUPLED\_COMPUTE: M3.5 native runner MERGED + FIRST REAL GPU RUN DONE on GCP L4 (M1? M3.5? M2?). VM deleted (clean slate). Resume/repro from `deploy/gcp/README.md` + `docs/DECOUPLED\_COMPUTE/DEPLOY\_REPORT\_GCP\_2026-06-20.md`.\*\* Resumes the engine-side dependency the appliance block above was waiting on.

34 - ? \*\*M3.5 MERGED ? [#256](https://github.com/avidullu/badminton-highlight-indexer/pull/256):\*\* `NativeWasbRunner` (`backend/pipeline/detectors/native\_wasb\_runner.py`) runs `wasb\_infer.py` NATIVELY on a Linux GPU box ? no WSL/conda-activate/mnt; `detector\_impl="native"` seam; `device` cuda/mps/cpu (cuda default = byte-parity with WSL); weights/repo/python/scratch env-first

(`WASB\_WEIGHTS`/`WASB\_REPO\_DIR`/`WASB\_PYTHON`/`WASB\_SCRATCH`). `wasb\_infer.py` got `--device` + device-keyed cache (back-compat). \*\*`worker train --trajectory-source detector`\*\* does REAL per-video extraction (fps PROBED via ffprobe + skip-on-miss, video-ext filter, graceful native-misconfig). Suite \*\*998 green\*\*, ruff+mypy clean, cov 83.7%. Adversarial multi-agent review run; fixes folded in. \*\*Parity ? VALIDATED 2026-06-20 on the owner's RTX 2070 (via WSL ? a Linux+CUDA proxy): native==WSL 1194/1195 frames, two native runs byte-identical (deterministic); gate = ~1e-3px TOLERANCE (cuDNN sub-pixel cross-GPU), NOT byte-exact. Native runner now DEFAULTS to streaming (#259). See the validation checkpoint below.\*\*

35 - ? \*\*DO?GCP PIVOT (ADR-010) + runbook ? [#257](https://github.com/avidullu/badminton-highlight-indexer/pull/257) (docs):\*\* DO has NO Singapore/India GPU (NA-only droplets) ? GPU+bucket move to \*\*GCP asia-south1 (Mumbai)\*\*. `deploy/gcp/{README,provision.sh,teardown.sh}` = full runbook + live-resource inventory + switch-off. \*\*PROVISIONED LIVE\*\* (project `gen-lang-client-0306026826`, billing \*\*Khelsutra Billing\*\* `01420D-419EE0-53D83C`): all APIs on . bucket \*\*`gs://khelsutra-rally-corporus`\*\* (asia-south1) . SA \*\*`rally-worker`\*\* + key at \*\*`~/gcp/rally-worker-sa.json`\*\* (local only, gitignored) . \*\*`\$100` budget `rally-100usd-guard`\*\* (50/90/100%).

36 - ? \*\*FIRST REAL GPU RUN DONE on GCP (2026-06-20) ? report

```
[#261](https://github.com/avidullu/badminton-highlight-indexer/pull/261)
`docs/DECOUPLED_COMPUTE/DEPLOY_REPORT_GCP_2026-06-20.md`.** GPU quota GRANTED
(`GPUS_ALL_REGIONS`?1, via Cloud Quotas REST API ? **Spot does NOT bypass it, tested**). Spun an
L4 (`g2-standard-4`, asia-south1-c; T4 was stocked-out in -a; image `common-
cu129-ubuntu-2204-nvidia-580`), ported via `bootstrap --gpu` + the FIXED `gpu_setup.sh`. **M1
? ** `worker infer detector_impl=native` on the smoke clip ? **22 rallies** (gcs in?out);
highlight reel stitched ? `gs://khelsutra-rally-
corpus/outputs/05e0e577?/highlight_smoke_l4.mp4`. **M3.5 ? ** native runner on a real cloud GPU;
**cross-GPU parity = sub-pixel (684/1195 frames differ, cuDNN L4-vs-2070) ? gate at the RALLY
level, not CSV bytes** (same-GPU is byte-exact). **M2 ? ** real-extract train validated IN-FLIGHT
(16k+/21k windows extracted on the L4) ? full weights = next session. **VM DELETED** (clean
slate, $0; bucket persists). **Ops Agent fixed** (root cause: project's **logging+monitoring
APIs were OFF** ? now enabled + SA monitoring/logging roles + agent installed).
37 - **BUCKET state** `gs://khelsutra-rally-corpus`: `models/wasb_badminton_best.pth.tar` .
`labels/*.rallies.csv` (7) . `videos/` (smoke + mahadevpura-1 + GX030094_proxy) .
`outputs/05e0e577?/{intervals.csv,report.json,highlight_smoke_l4.mp4}` .
`outputs/parity/native_smoke_l4.csv`.
38 - **NEXT ITEMS (next session picks up):** (1) **Finalize M2** ? re-port one-shot via the
fixed `gpu_setup.sh` (repro: `deploy/gcp/README.md`), run `worker train --trajectory-source
detector` with **`indexing.wasb.stream_video=false`** (disk extraction keeps the GPU fed on LONG
videos ? streaming starved it; finding #6) ? weights `weights/0.1.0-l4/`. (2) **Full corpus**
for real training: gdown the 4 big videos + held-out Video 1 from the Drive folder
`1sHGL7iacnmM80ChT-p2cKy638gK504T_` ? bucket `videos/` (raw GoPro at `F:\[Khelsutra] GoPro Hero
Black 12`). (3) **D0 GPU is BLOCKED at the provider** ? token valid but **0 GPU Droplet sizes on
the account** ; enable GPU Droplets in the D0 console to use D0 credits for training (else
training stays on GCP). (4) D0 cleanup still owner-side: copy `/srv/khelsutra-site` off droplet
`579001481` ? destroy it ? rotate the exposed Spaces key. (5) Rally-level cross-GPU parity check
+ the streaming-vs-disk train default. PRs this session: **#256?#261** all merged.
[[monetization-plan]] [[working-agreement-merging]] [[paper-coauthor-aim]]
39
40 **Checkpoint:** 2026-06-20 late (NEW SESSION ? D0 still BLOCKED ? M2 finalize on a GCP
L4; **ADR-012** GPU/Cloud-Run RATIFIED; ops-agent fixed+made-permanent; per-video workspace
kicked off.) Repo is now under the **`Khelsutra`** org (`github.com/Khelsutra/badminton-
highlight-indexer`).
```

## 69. assistant (2026-06-21T00:48:23.119Z)

This is directly the M2/GCP-L4 provenance follow-up. Let me add a concise checkpoint line after the DECOUPLED\_COMPUTE block's checkpoint.

## 70. user (2026-06-21T00:48:33.178Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

## 71. assistant (2026-06-21T00:48:38.561Z)

Now let me check CI status on the PR. Per the merging working-agreement, I can merge a good-engineering fix on verified-green CI.

## 72. user (2026-06-21T00:48:47.210Z)

```
Mypy (lenient baseline) pending 0 https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27888857806/job/82528681302
Golden fixtures cross-OS (macos-latest) pass 22s https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/27888857806/job/82528681274
Ruff lint pass 8s https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27888857806/job/82528681270
Syntax sweep + import smoke (no GPU stack) pass 18s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27888857806/job/82528681271
Full test suite (offline, mocked) pending 0 https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/27888857806/job/82528681357
```

```
Golden fixtures cross-OS (ubuntu-latest) pass 23s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27888857806/job/82528681308
Golden fixtures cross-OS (windows-latest) pending 0
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27888857806/job/82528681276
--- view ---
{"mergeState":"UNSTABLE","mergeable":"MERGEABLE","state":"OPEN"}
```

### 73. assistant (2026-06-21T00:48:54.280Z)

CI is partway: ruff, syntax sweep, golden fixtures (mac/ubuntu) green; mypy, full suite, windows fixtures pending. Let me watch until they finish.

### 74. user (2026-06-21T00:50:40.407Z)

```
Mypy (lenient baseline) pass 34s https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27888857806/job/82528681302
Ruff lint pass 8s https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27888857806/job/82528681270
Syntax sweep + import smoke (no GPU stack) pass 18s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27888857806/job/82528681271
Full test suite (offline, mocked) pending 0 https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/27888857806/job/82528681357
Golden fixtures cross-OS (ubuntu-latest) pass 23s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27888857806/job/82528681308
Golden fixtures cross-OS (windows-latest) pass 52s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27888857806/job/82528681276
Full test suite (offline, mocked) pass 2m10s https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/27888857806/job/82528681357
Golden fixtures cross-OS (macos-latest) pass 22s https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/27888857806/job/82528681274
Golden fixtures cross-OS (ubuntu-latest) pass 23s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27888857806/job/82528681308
Golden fixtures cross-OS (windows-latest) pass 52s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27888857806/job/82528681276
Mypy (lenient baseline) pass 34s https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27888857806/job/82528681302
Ruff lint pass 8s https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27888857806/job/82528681270
Syntax sweep + import smoke (no GPU stack) pass 18s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27888857806/job/82528681271
Full test suite (offline, mocked) pass 2m10s https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/27888857806/job/82528681357
Golden fixtures cross-OS (macos-latest) pass 22s https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/27888857806/job/82528681274
Golden fixtures cross-OS (ubuntu-latest) pass 23s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27888857806/job/82528681308
Golden fixtures cross-OS (windows-latest) pass 52s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27888857806/job/82528681276
Mypy (lenient baseline) pass 34s https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27888857806/job/82528681302
Ruff lint pass 8s https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27888857806/job/82528681270
Syntax sweep + import smoke (no GPU stack) pass 18s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/27888857806/job/82528681271
```

```
=== final mergeState ===
{"mergeState":"CLEAN","mergeable":"MERGEABLE","state":"OPEN"}
```

#### 75. assistant (2026-06-21T00:50:59.972Z)

?? The working-tree snippets show my harness/test edits reverted to the original ? a sign the **\*\*shared checkout was switched or reset by a sibling session\*\***. My commit is already pushed (PR #269, CI green on commit `fc862a9`), so the PR is unaffected ? but let me verify the local git state carefully without touching anything.

#### 76. user (2026-06-21T00:51:03.395Z)

```
=== current branch + HEAD ===
docs/archive-decoupled-v2
fb51cdc docs: archive DECOUPLED_COMPUTE ? DELIVERED (M0-M2 + M3.5 on GCP)

=== is my commit fc862a9 reachable / on remote branch? ===
fc862a9 fix(gen0): provenance-stamp the manifest git_sha off a git-archive tarball
e6110ce feat(api): GET /api/health + return download_url from /api/compile (M1/P6) (#268)
a555887 docs(deploy): M2 FINALIZED ? Gen-0 weights/0.1.0-l4 pushed; both VMs deleted ($) (#267)

=== does my pushed commit contain the new _git_sha? ===
58:GIT_SHA_ENV = "RALLY_GIT_SHA"
61:def _valid_sha(s: Optional[str]) -> Optional[str]:
74: 2. the ``RALLY_GIT_SHA`` env var ? out-of-band injection for a run off a ``git archive``
84: live = _valid_sha(out)
89: env_sha = _valid_sha(os.environ.get(GIT_SHA_ENV))
94: file_sha = _valid_sha(f.read())

=== working-tree harness right now ===
50:def _git_sha() -> str:
```

#### 77. assistant (2026-06-21T00:51:28.166Z)

Now it's clear and **\*\*no work was lost\*\***: my commit `fc862a9` is safely on `origin/fix/gen0-git-sha-provenance` with all changes intact, and PR #269 passed CI. A sibling session simply switched the shared checkout to its own new branch `docs/archive-decoupled-v2` (committing the DECOUPLED\_COMPUTE doc-archive moves I saw staged earlier) ? that's why my working-tree files **\*appear\*** reverted. I'm now on the sibling's branch, so I'll **\*\*leave the working tree untouched\*\*** (they're actively using it).

CI is fully green and the PR is `CLEAN`/`MERGEABLE`. This is a good-engineering fix (provenance correctness ? not product/strategy/licensing), which I'm authorized to merge on verified-green CI. My changed files are disjoint from the sibling's (no conflict). Let me merge via squash.

#### 78. user (2026-06-21T00:51:32.473Z)

```
=== final pre-merge gate ===
 1 pass Full test suite (offline, mocked)
 1 pass Golden fixtures cross-OS (macos-latest)
 1 pass Golden fixtures cross-OS (ubuntu-latest)
 1 pass Golden fixtures cross-OS (windows-latest)
 1 pass Mypy (lenient baseline)
 1 pass Ruff lint
 1 pass Syntax sweep + import smoke (no GPU stack)
--- mergeable ---
{"mergeState":"CLEAN","mergeable":"MERGEABLE","state":"OPEN"}
```

#### 79. assistant (2026-06-21T00:51:35.000Z)

All 7 checks pass; `CLEAN`/`MERGEABLE`. Merging via squash and deleting the remote branch.

#### 80. user (2026-06-21T00:51:40.624Z)

```
=== confirm merged ===
{"mergedAt":"2026-06-21T00:51:38Z","state":"MERGED"}
```

### 81. assistant (2026-06-21T00:51:48.596Z)

Merged at 2026-06-21T00:51:38Z. Let me verify it landed on `master` and update the memory checkpoint to reflect the merge.

### 82. user (2026-06-21T00:51:52.610Z)

```
- [deleted] (none) -> origin/fix/gcp-teardown-default-delete
- [deleted] (none) -> origin/fix/gen0-git-sha-provenance
 e6110ce..c8e60ee master -> origin/master
=== master tip now carries the fix? ===
c8e60ee fix(gen0): provenance-stamp the manifest git_sha off a git-archive tarball (#269)
e6110ce feat(api): GET /api/health + return download_url from /api/compile (M1/P6) (#268)
a555887 docs(deploy): M2 FINALIZED ? Gen-0 weights/0.1.0-l4 pushed; both VMs deleted ($) (#267)
--- _git_sha on origin/master ---
2
```

### 83. assistant (2026-06-21T00:51:56.345Z)

The fix is on `origin/master` as `c8e60ee` (#269), the remote branch is pruned, and `RALLY\_GIT\_SHA` appears in the merged harness. Let me update the memory checkpoint from "OPEN" to "MERGED".

### 84. user (2026-06-21T00:52:01.167Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

### 85. assistant (2026-06-21T00:52:18.047Z)

Done. Here's the summary.

## ? Shipped ? PR [#269](https://github.com/Khelsutra/badminton-highlight-indexer/pull/269) merged to `master`

**\*\*Root cause:\*\*** a `git archive` tarball carries no `.git`, so `\_git\_sha()` in [training/gen0/harness.py](training/gen0/harness.py:50) couldn't run `git rev-parse` on the remote box and stamped `git\_sha:"unknown"` ? which is why `gs://khelsutra-rally-corpus/weights/0.1.0-l4/manifest.json` shipped with no real provenance.

**\*\*Fix\*\*** ? `\_git\_sha()` now resolves the SHA in priority order, hex-validated at each step:

1. **\*\*live `git rev-parse HEAD`\*\*** ? local dev / CI; authoritative, **\*\*byte-identical to before\*\*** (default-OFF-safe)
2. **\*\*`\$RALLY\_GIT\_SHA`\*\*** env var ? out-of-band injection for a tarball run (auto-inherited by the harness subprocess the worker spawns)
3. a **\*\*`GIT\_SHA`\*\*** file baked into the repo root at tarball-build time
4. **\*\*`"unknown"`\*\*** ? last resort, never fabricated

A `^[0-9a-f]{7,40}$`` guard means a stray value (e.g. a literal `"unknown"`) can never be stamped as a real SHA.

**\*\*Changes (4 files):\*\*** harness resolution chain · [deploy/gcp/README.md](deploy/gcp/README.md) \$4 now bakes the SHA into the archive (`git archive --add-file=/tmp/GIT_SHA``) or `export RALLY_GIT_SHA`` on the box · `/GIT_SHA`` gitignored · **\*\*6 new unit tests\*\*** (priority, env/file fallbacks, garbage-rejection, and end-to-end manifest carrying a real SHA off a simulated tarball).

**\*\*Verified locally:\*\*** `run_all_tests.py`` ? **\*\*1032 passed\*\***; `ruff`` + `mypy`` clean; harness coverage 93% (CI floor 70%). **\*\*CI:\*\*** all 7 checks green before merge.

## Two things to flag

1. **\*\*Forward-looking only\*\*** ? the already-shipped ``0.1.0-l4`` manifest still reads ``"unknown"``. Re-stamping it needs an owner-gated re-run (out of scope here).
2. **\*\*Shared-checkout collision (handled, no work lost).\*\*** While I worked, a sibling session had staged ``docs/DECOUPLED_COMPUTE`` ? `docs/archives/past_projects/?`` renames into the shared index, then switched the checkout to its own branch ``docs/archive-decoupled-v2``. I committed via **\*\*explicit pathspec\*\*** so only my 4 files landed (their staged renames untouched), and my commit was already pushed + merged independently of the local working tree. The checkout is currently on the sibling's branch ? I **\*\*left it untouched\*\***. This is exactly the `[[gotcha-shared-checkout-branch-collisions]]` scenario; the protocol held.

### 86. user (2026-06-21T00:52:20.468Z)

merge the PR when tests and CI are green

### 87. assistant (2026-06-21T00:52:24.743Z)

PR #269 is already merged ? that happened in the previous turn once all CI went green. Let me confirm the current state so I'm reporting accurately.

### 88. user (2026-06-21T00:52:26.917Z)

```
{"mergeCommit": "c8e60ee18d87f9aee44c8520b347da58a21113ef", "mergedAt": "2026-06-21T00:51:38Z", "number": 269, "state": "MERGED"}
--- master tip ---
c8e60ee fix(gen0): provenance-stamp the manifest git_sha off a git-archive tarball (#269)
```

### 89. assistant (2026-06-21T00:52:33.216Z)

Already done ? there's nothing left to merge.

PR [#269](<https://github.com/Khelsutra/badminton-highlight-indexer/pull/269>) is **\*\*MERGED\*\*** (squash commit ``c8e60ee``, merged 2026-06-21T00:51:38Z), and it's the current tip of ``origin/master``. I merged it last turn only after confirming all 7 CI checks were green (``Full test suite``, ``Ruff lint``, ``Mypy``, ``Syntax sweep``, and ``Golden fixtures`` across macOS/Ubuntu/Windows) and the PR was ``CLEAN`/`MERGEABLE``. The remote branch was auto-deleted on merge.

So your condition ? merge when tests and CI are green ? was satisfied and the merge is complete. No further action needed.